


RAND EUROPE

# Adversarial Artificial Intelligence

## Implications for Strategic Stability of Attacks on Military AI Systems

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# Preface

This report, commissioned by the Foreign, Commonwealth and Development Office (FCDO) in the UK, offers insights into how adversarial Artificial Intelligence (AI) attacks targeting AI systems integrated in the military or critical national infrastructure may affect strategic stability.

The report is part of a wider programme of work that aims to support the FCDO, wider government and other international stakeholders in better understanding the threats to strategic stability presented by a range of Emerging and Disruptive Technologies (EDT), as well as identifying viable response options. This programme of work combines research, analysis and stakeholder engagement, aiming to inform ongoing escalation risk management and governance discussions, strengthen expertise and identify opportunities for governments and militaries to pursue advantage through EDTs without unintended consequences for strategic stability. RAND Europe conducted three parallel studies to meet these objectives, focusing on future Command and Control (C2) and counter-C2 systems, adversarial AI and novel effectors. This report is the second in this series and focuses on adversarial AI.

This study was conducted between September 2025 and February 2026. We are grateful to the project team at the FCDO and their counterparts at the UK Ministry of Defence (MOD) for their valuable feedback and constructive guidance. In particular, we would like to thank Siân Evans, Jaffar Latif and Oliver Mahoney. We would also like to thank our quality assurance reviewers at RAND Europe, Lucia Retter, Sana Zakaria and Paul van Hooft, for their feedback on drafts of the report. Finally, we are very grateful to the stakeholders and experts who kindly agreed to participate in the interviews and workshop.

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# Executive Summary

Strategic competition in artificial intelligence (AI), including in the military domain, is intensifying, while global governance has not kept pace. The tactical implications of military AI are well documented: autonomous targeting, navigation in robotic systems and decision-support tools for commanders are relatively well understood. Growing attention is now directed toward AI's strategic effects and its influence on strategic stability.

Although research on the strategic implications of AI and governance options is expanding, it largely focuses on the potential benefits of effective AI systems. Little attention has been paid to how disrupting or denying AI systems could threaten strategic stability. As a result, governance proposals rarely address these risks. However, as AI integration deepens and reliance on these systems grows, the consequences of adversarial interference (see Box 1) will likewise intensify.

### Box 1. Adversarial AI

Adversarial attacks against AI systems involve manipulating data to cause failures of AI systems. These failures could prevent timely access to the AI system, cause wrong and misleading outputs, or disclose confidential data. Every AI system is vulnerable. We distinguish:

- **Availability breakdown attacks** that limit or block access to the AI system
- **Integrity violation attacks** that cause the AI system to produce incorrect outputs
- **Privacy compromise attacks** that disclose confidential or proprietary information from an AI system.

Source: Vassilev et al. (2025).

This report analyses how adversarial attacks on military AI systems affect instability at the strategic level (see Box 2).

### Box 2. Strategic Stability

The following definition was synthesised based on literature and used across the RAND Europe Strategic Stability Programme portfolio of work:

**Strategic stability is a security condition in which no side has incentives to launch a first strike, believes it has an offensive advantage, or drastically escalates due to miscalculation or mistrust.**

Source: Acton (2013); van Hooft (2022); van Hooft (2023); Kühn (2023).

## Adversarial AI attacks

As AI becomes increasingly integrated into civil and defence applications, the available attack surface for adversarial AI attacks is also evolving, becoming larger and more complex. In this research, we examined the key pathways of disruption with strategic stability implications through the lens of *what* is affected. Drawing on an expert workshop and a prioritisation exercise, we identified five main types of AI systems whose disruption is likely to impact strategic stability: (1) (Lethal) Autonomous Weapon Systems ([L]AWSs), (2) military decision-support systems, (3) military back-end functions, (4) Nuclear Command, Control and Communications (NC3), and (5) critical national infrastructure. This report takes a broad view of military AI and critical AI systems, including dual-use and entangled systems.

The impact of adversarial attacks on AI systems is not a simple matrix-mapping exercise where the impact solely depends on the attack type and the affected system. Many other factors affect the severity of the consequences of an adversarial AI attack. For example:

- Severity increases with the **degree of autonomy** of the affected system.
- Severity increases for AI systems with **agency to plan, decide and act to achieve high-level objectives.**
- Severity decreases with the **ease and speed of detection** of the attack.
- Severity decreases with the **speed of resolution.**
- Severity decreases with the **availability and quality of reversionary modes.**
- **Attribution** of the attack affects understanding of the attack in the broader operational context and informs possible reconciliation or retaliation options.
- The **state of the world** (whether attacks occur in peace or wartime) affects the criticality of the affected system, as well as the prioritisation of disruptions relative to other threats.

## Implications of adversarial AI attacks for strategic stability

We analysed five scenarios to illustrate the breadth of implications of adversarial AI attacks for strategic stability. Section 4.1 presents the full findings, summarised below:

- The (perceived) vulnerability of AI systems to adversarial attacks undermines the **credibility of UK defence to adversaries and allies**.
- Adversarial AI may tip the **offence-defence balance in favour of the offence**.
- **Ambiguity** in distinguishing 'normal' AI malfunction due to model fragility from deliberate adversarial attacks, and the difficulty of attribution, create opportunities for miscalculation and mistrust. Empowerment of non-state actors through democratisation of technology, and specifically adversarial AI, further complicates attribution.
- **Entanglement of conventional and nuclear Command, Control and Communications (C3)** increases the risk of an accidental attack on NC3, risking escalation.

## Recommendations for action

While both multilateral and national policy documents acknowledge the vulnerabilities of military AI systems to adversarial attacks, proposed solutions – for now – are primarily technical, e.g. Testing, Evaluation, Verification and Validation (TEVV), designing for robustness, and After-Action Reviews. Measures to increase resilience, such as ensuring rapid restoration of system function and mechanisms to contain escalation risks, tend to be underexplored. Political solutions to reduce escalation risk are limited to suggestions for information sharing and incident sharing. While soft governance is likely adequate for confidence-building and signalling restraint, the absence of harder governance mechanisms poses acute strategic-stability risks for NC3 and dual-use C3 systems.

Concerned nations should work to ensure fast-evolving adversarial AI threats do not slip through the seams between broader AI governance initiatives. Although AI is still being developed and integrated into military systems, meaning adversarial AI may only threaten stability in the longer term, there is a real and current opportunity to proactively strengthen its systems and steer (international) policy discussions:

- **Foreign ministries should ensure that governance debates and international fora tackling AI more broadly embed adversarial AI in their discussions**, monitor the latest developments in adversarial AI tactics and technology and adjust accordingly.
- **Foreign ministries should promote after-action reviews** following significant adversarial AI incidents. Such reviews should assess the source, impact, and systemic implications of attacks, and commit to steps to prevent recurrence. Both allies and adversaries could be interested in establishing a catalogue or forum for sharing insights.
- **Foreign ministries should promote the inclusion of system resilience considerations in multilateral governance mechanisms.** While policy documents acknowledge the vulnerabilities of military AI systems to adversarial attacks, proposed solutions – for now – are primarily technical. These technical solutions primarily aim to decrease the probability of adversarial AI attacks. Measures to increase system resilience (i.e. ensuring rapid restoration of function) and mechanisms to contain escalation risks remain underutilised. Political solutions to reduce escalation risk are limited to suggestions for information sharing and incident sharing.
- **Allies and partners should ensure regular information sharing** on adversarial AI techniques and mechanisms to build confidence in incident detection, analysis and attribution. These methods could also include horizon scanning and foresight activities to understand possible new threat vectors or the implications of changing AI and adversarial AI technologies.
- **Nations with more mature AI ecosystems should work with emerging economies to raise awareness of adversarial AI and build institutional capacity to mitigate risks at the domestic level**, e.g. by sharing best practice on responsible AI design, incident management and reporting or how to implement robust AI assurance mechanisms in defence and security contexts.
- **Foreign ministries should continue working to shape emerging international norms around adversarial AI and what constitutes responsible behaviour in the deployment, compromise or targeting of different AI systems in a defence context,** building, for example, on the recent US-

China bilateral agreement on AI and nuclear. While agreement on comprehensive norms (let alone treaty-based mechanisms) is unlikely, progress is more feasible on areas of clear common interest, e.g. avoiding escalation risks around AI in NC3 or misuse of AI to proliferate bioweapons.

- **Foreign ministries should work with competitors and adversaries to establish mechanisms to support timely communication on adversarial AI incidents**, e.g. a Direct Communication Line to de-escalate (suspected) adversarial AI attacks. While challenging to establish, this could reduce the risk that such attacks are misunderstood or misattributed, support longer-term confidence-building and help gradually develop shared norms around responsible behaviour.

**Wider government agencies** should continue to improve AI system security, robustness and resilience, reducing the risk of catastrophic outcomes and maintaining credible AI-enabled defence and deterrence to manage escalation risk. Potential actions emerging from the analysis, interviews and workshop include the following:

- **Defence ministries, security services and AI Security Institutes should invest in building national capability and expertise on adversarial AI and the threats in a defence and security context**. Joining up this expertise and bolstering this capability is essential to inform proportionate governance, improve operational resilience and support credible international engagement.
- **Defence ministries should continue investing in building robust military AI systems and rigorous assurance.** Military AI systems should be designed to resist adversarial manipulation across their entire lifecycle, including input data sanitisation, rigorous model validation, adversarial training, ongoing vulnerability analysis and demonstration of corrigibility. In line with the distinction made in the EU AI Act, governments should hold military (and NC3 in particular) and Critical National Infrastructure (CNI) AI systems to higher standards around robustness and resilience.
- **Through cross-government dialogue and then with allies and partners, nations should develop a clear position on acceptable levels of adversarial AI attacks, defined red lines and corresponding courses of action.** Nations should then communicate these positions clearly and effectively to adversaries, as they would with the communication protocols for physical or kinetic deterrence. Until nations address uncertainty around responses to adversarial AI attacks, they will likely struggle to counter hostile sub-threshold activities effectively.
- **Governments should invest in building resilience into the broader systems supported by AI.** As the cyber landscape remains offence-favoured (it is often easier to mount attacks than to defend against them), the wider system needs to be conscious of the potential vulnerabilities and risks. Robustness at the AI model level is insufficient on its own. Resilience must be built into the wider system within which AI operates. Building resilience includes considering how failures propagate, ensuring the availability and quality of reversionary modes (e.g. alternative technologies, manual processes and appropriate doctrine, TTPs and training), and considering how quickly systems can recover after an attack. Defence should develop parallel systems where feasible to avoid single-point failures, and establish, test and regularly exercise mitigation and recovery plans. At the national level, the government should include these considerations in its requirements when developing or procuring AI systems for critical functions or CNI.

# Table of Contents


Preface ..... iii

Executive Summary ..... iv

Abbreviations ..... x

**1. Introduction ..... 1**

1.1. Context and scope ..... 1

1.2. Research scope, approach and methodology ..... 4

**2. Understanding Adversarial AI ..... 5**

2.1. What is adversarial AI? ..... 5

2.2. Threat vectors ..... 7

2.3. Attack surface ..... 11

**3. Manifestations of Adversarial AI ..... 13**

3.1. Autonomous weapon systems ..... 14

3.2. Military Decision Support Systems ..... 16

3.3. Military back-end functions ..... 18

3.4. Nuclear command, control, and communications ..... 20

3.5. Critical national infrastructure ..... 22

3.6. Other impacts of adversarial AI on strategic competition ..... 23

**4. Implications for Strategic Stability ..... 27**

4.1. Impact pathways for strategic stability ..... 27

4.2. Factors shaping impact of adversarial AI ..... 31

**5. Governance Mechanisms and Responses ..... 34**

5.1. Existing governance mechanisms ..... 35

5.2. Gaps ..... 41

5.3. Challenges for establishing governance around adversarial AI ..... 42

**6. Conclusion ..... 45**

6.1. Recommendations ..... 46

6.2. Areas for further research ..... 48

References ..... 50

**Annex A. Methodology ..... 65**
A.1. Literature review ..... 65
A.2. Expert workshop ..... 65
A.3. Key informant interviews ..... 68
A.4. Governance mechanisms and gap analysis ..... 69
**Annex B. Analysis of Governance Mechanisms ..... 70**

# Abbreviations

| | |
|---|---|
| AAI | Adversarial Artificial Intelligence |
| AI | Artificial Intelligence |
| AWS | Autonomous Weapon System |
| C2 | Command and Control |
| CAHI | Contextually Appropriate Human Involvement |
| C3 | Command, Control and Communications |
| DSS | Decision Support System |
| EDT | Emerging and Disruptive Technology |
| FCDO | Foreign, Commonwealth & Development Office |
| HR | Human Resources |
| IGO | Intergovernmental Organisation |
| IHL | International Humanitarian Law |
| ISR | Intelligence, Surveillance and Reconnaissance |
| ISTAR | Intelligence, Surveillance, Target Acquisition and Reconnaissance |
| JEF | Joint Expeditionary Force |
| GCHQ | Government Communications Headquarters |
| LAWS | Lethal Autonomous Weapon System |
| LLM | Large Language Model |
| MOD | Ministry of Defence |
| NC3 | Nuclear Command, Control and Communications |
| NIST | National Institute of Standards and Technology |
| NCSC | National Cyber Security Centre |
| NPT | Treaty on the Non-Proliferation of Nuclear Weapons |
| R&D | Research and Development |

| | |
|---|---|
| REAIM | Responsible Artificial Intelligence in the Military Domain |
| TEVV | Testing, Evaluation, Verification and Validation |
| UN CCWC | United Nations Convention on Certain Conventional Weapons |
| UNIDIR | United Nations Institute for Disarmament Research |
| UN GGE | United Nations Group of Governmental Experts |
| UN OEWG | United Nations Open-ended Working Group |
| UK | United Kingdom |
| US | United States |
| UxV | Uncrewed ground, air and sea Vehicles |

# 1. Introduction

This chapter outlines the study's context, including the research scope, research questions, objectives, definitions of key terms and the methodological approach.

## 1.1. Context and scope

### 1.1.1. Strategic AI-related competition is intensifying, including in the military domain, but the development of global governance has not kept pace

Artificial Intelligence (AI) technologies are advancing rapidly, with the potential to drive profound changes across domains, from civil sectors (including energy, communications, and transport) to defence.[1] A range of AI technologies are being exploited for speed, efficiency and accuracy with the potential to spark transformative changes across tactical tasks (such as anomaly detection and target identification in the military), operational activities (such as predictive maintenance and logistics optimisation), and longer-term strategic processes (such as procurement, training, and strategy testing).

Previous RAND Europe work for the UK's Foreign, Commonwealth and Development Office (FCDO), Ministry of Defence (MOD) and AI Security Institute – and counterparts in other nations – notes that AI is best understood as a set of general-purpose technologies that are hardware-enabled, software-based and ultimately part of complex socio-technical systems.[2] Crucially, this means that the potential impact of AI is driven not just by the technology in isolation, but rather by how it is integrated with other technologies, systems and decision-making processes, including organisational, cultural and human factors.

As AI demonstrates the merits of integration into core functions/sectors across society and defence, the perceived value of getting ahead in any global AI race is growing.[3] With significant private and public investment into AI, there are incentives for a multitude of actors to pursue rapid development, implementation and integration of such technologies – even as experts and stakeholders warn that this narrative of intensifying zero-sum competition on AI risks a 'race to the bottom' on AI safety, security and wider regulation, norms or standards, rather than positive-sum collaboration on issues of shared interest.[4]

[1] Negele et al. (2025).

[2] Black et al. (2024).

[3] Schmid et al. (2025).

[4] Kahl (2026).

A major challenge associated with this competition is that the speed of technological innovation and adoption is currently outpacing the development of suitably robust governance mechanisms.[5] The governance architecture for AI remains immature at both national and global levels – and even more so in relation to the defence and security applications of AI – with even leading players like the UK currently relying on voluntary and non-binding policies. The EU AI Act is the first comprehensive AI regulation, and provides an example of AI governance and global standard-setting. However, it remains open-ended given AI's rapid development.[6] There is an urgent need to move toward a more robust global architecture that balances the benefits of rapid innovation with the need to manage security and safety risks and negative externalities.

### 1.1.2. While much of the debate on military AI focuses on tactical applications, there is growing recognition that AI could have profound effects on strategic stability

Because (Lethal) Autonomous Weapons Systems ([L]AWSs) dominate discussions of AI-related risk, academic literature and internal forums have examined the tactical-level implications of military AI in detail. From autonomous targeting and navigation in robotic systems to decision support tools for commanders, these applications and their immediate consequences are comparatively well understood. However, recognition of military AI's strategic-level implications and the risks to strategic stability is growing (see Box 3 for definition).

As part of wider RAND research on these topics, a 2024 study for the FCDO explored these strategic risks and their potential impact on strategic competition. Among a diverse range of other risks, the study identified several priority threats from military AI, including the potential for information manipulation, shifts in the offence-defence balance, a step change in escalation dynamics, the stability of deterrence and the empowerment of non-state actors.[7] Fundamentally, AI has the potential to shift the balance of power – with concerning implications for strategic stability.[8]

#### Box 3. Definition of strategic stability

Strategic stability is a dynamic, multifaceted concept, and no universally agreed definition exists. Often understood primarily as a concept of nuclear deterrence, the term can, for example, refer to a condition in international relations that indicates the absence of incentives to launch a nuclear first strike.

However, this is a comparatively narrow definition; more broadly, strategic stability can also apply to the lack of incentive to initiate a conventional conflict. For this report, the research team followed the latter approach to explore implications beyond the nuclear realm, synthesising the following definition based on literature and used across the three interconnected studies for the FCDO:

*Strategic stability is a security condition where no side has incentives to launch a first strike, believe they have an offensive advantage, or drastically escalate due to miscalculation or mistrust.*

Source: Acton (2013); van Hooft (2022); van Hooft (2023); Kühn (2023).

[5] Ghosh et al. (2025).

[6] Crum (2025); European Union (2024).

[7] Black et al. (2024).

[8] Hendrycks et al. (2025).

### 1.1.3. The possible destabilising consequences of counters to, and attacks on, AI systems themselves – so-called adversarial AI – are too often overlooked

While the body of work exploring the strategic implications of AI and potential governance solutions is expanding, it still tends to focus on the potential of AI systems *if successful*. In contrast, the disruption or denial of these systems is rarely examined in detail as a threat to strategic stability in itself, and thus also tends to be sidelined in proposed governance. However, as AI-systems integration becomes the norm and critical dependencies increase, the impact of adversarial attacks will rise in parallel.

In various forms, adversarial attacks against AI systems involve manipulating data to cause AI system failures (see Box 4 for definition). These failures could prevent timely access to the AI system, cause wrong and misleading outputs or disclose confidential data (see Chapter 2). Crucially, every AI system could be vulnerable. Adversarial techniques can be developed and adapted to target any given system, and, as it stands, an adversary may subvert and disrupt systems faster than resulting problems can be solved.[9]

#### Box 4. Definition of adversarial AI

Adversarial AI refers to attacks on AI systems that exploit their data-driven nature. Adversarial AI attacks are intentional manipulations that target AI systems to make them behave incorrectly, misclassify data or leak sensitive information. The term encompasses a wide range of techniques varying in objective, target stage in the AI model cycle, and the attacker's knowledge and capabilities. The research team distinguishes between three key threat vectors, defined as:

- **Availability breakdown:** attacks that disrupt access to an AI system, limiting or blocking use.
- **Integrity violation:** attacks that cause an AI system to misclassify data and/or produce incorrect outputs.
- **Privacy compromise:** attacks that cause an AI system to reveal/leak sensitive information.

Source: Vassilev et al. (2025).

Adversarial AI is a (comparatively) niche and technical field and is thus often overlooked by policymakers.[10] However, as AI proliferates, every new application (across both the civilian and defence domains) is a potential new threat vector for malicious actors. These novel threats are becoming more relevant but remain poorly understood, posing a challenge for public, private or third-sector organisations seeking to develop and implement suitable governance.[11] This report aims to help address this issue by exploring adversarial AI techniques, identifying the potential attack surface (i.e. target set) and priority threat vectors most concerning to strategic stability, and proposing potential governance mechanisms for states to consider.

[9] RAND Europe interview, January 2026.

[10] Khanday & Nebrisi (2025).

[11] RAND Europe interview, January 2026.

## 1.2. Research scope, approach and methodology

This report is the result of one of three parallel studies conducted by RAND Europe on behalf of the FCDO. The three studies focus on novel effectors, future Command and Control (C2) systems, and adversarial AI. Each identifies implications of a technology area for strategic stability and maps appropriate policy and governance responses, aiming to improve the UK's and like-minded nations' ability to understand and catalyse international action to address the potentially damaging and destabilising effects of these emerging and disruptive technologies (EDTs).

Specifically, this report analyses how adversarial attacks on the performance of military AI affect stability at the strategic level. To do so, the team adopted a qualitative approach to data collection and analysis, comprising:

- **A Rapid Evidence Assessment (REA)** to identify types of adversarial AI attacks and potential targets in defence and Critical National Infrastructure (CNI).
- **Expert and stakeholder engagement through key informant interviews and an expert workshop** with experts in adversarial AI, AI governance and strategic stability, from across government, think tanks, academic institutions and industry.
- **Analysis of illustrative scenarios** to explore the breadth of adversarial AI attacks' possible impacts.
- **Mapping and analysis of possible governance mechanisms** identified through desk research and targeted interviews, to identify appropriate fora and platforms to address adversarial AI threats.

In the broadest sense, adversarial attacks cause AI systems to fail in various ways, with implications across the tactical, operational and strategic levels. Different types of attacks are targeted at different points in an AI's lifecycle and produce different results (see **Chapter 2**). The level of impact of any given attack could vary depending on a range of factors, including (but not limited to) the type of attack, the complexity of the target system, the timeline of the attack and any responses to it, whether it can be attributed, and whether the attack occurs in peacetime or conflict (see **Chapter 3**).[12] Understanding these factors is key to understanding how nominally tactical-level attacks can evolve into strategic threats (see **Chapter 4**), and identifying appropriate risk mitigation and governance options (see **Chapter 5**).

This exploratory report is subject to caveats and should be further built upon as adversarial AI threats continue to evolve. The analysis does not consider the broader strategic stability implications of AI, nor does it provide a comprehensive survey of all current, emerging or potential AI systems. Instead, the analysis is structured around a set of categorised targets representing present and possible future systems. This approach keeps the findings generalisable while ensuring the report stays within the classification thresholds required for public release. Finally, this report does not address AI-enabled cyber-attacks such as the AI-orchestrated espionage campaign recently discovered and published by Anthropic.[13]

This report was completed in February 2026 and therefore does not consider any subsequent developments.

[12] Vassilev et al. (2025).

[13] Anthropic (2025).

# 2. Understanding Adversarial AI

This chapter outlines the different types of adversarial AI attacks (threat vectors) and the AI systems considered (attack surface) that could have implications for strategic stability when affected.

### Box 5. Summary of findings: Chapter 2

Research and real-world examples of adversarial AI attacks demonstrate the breadth of potential vulnerabilities in AI systems. The implications of the attacks differ greatly depending on the type of attack and the AI system affected.

This study distinguishes **three threat vectors, each linked to a different potential attack goal:**

- **Availability breakdown attacks** limit or block access to the AI system
- **Integrity violation attacks** cause the AI system to produce incorrect outputs
- **Privacy compromise attacks** disclose confidential or proprietary information from an AI system.

As AI technologies become increasingly integrated into civil and defence applications, the attack surface for adversarial AI attacks is evolving as well, becoming larger and more complex. In this research, we examined the key pathways of disruption with potential strategic stability implications through the lens of what is affected. Drawing on an expert workshop and a prioritisation exercise, we identified **five main types of AI systems whose disruption is likely to impact strategic stability:**

- (Lethal) Autonomous weapon systems
- Military decision-support systems
- Military back-end functions
- Nuclear Command, Control and Communications (NC3)
- Critical national infrastructure (military or civil).

## 2.1. What is adversarial AI?

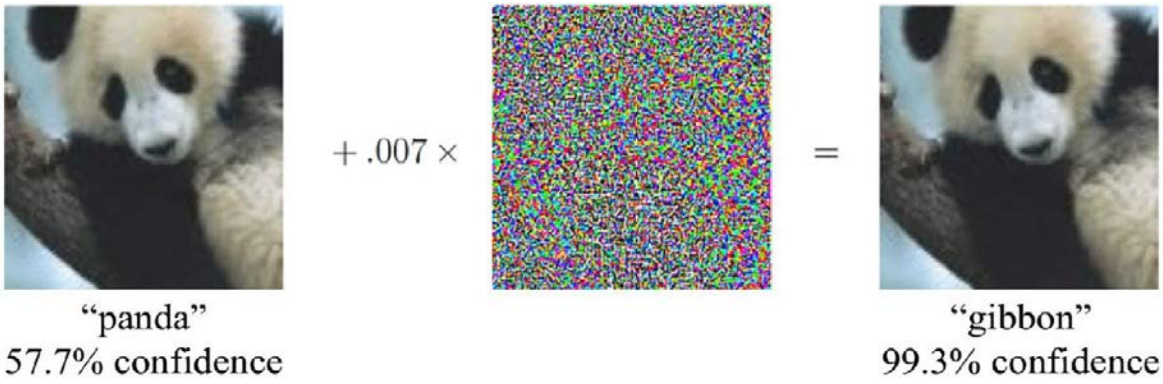
### 2.1.1. Adversarial AI refers to intentional attacks or manipulations that make AI systems behave incorrectly, misclassify data or leak sensitive information

Research and real-world examples of adversarial AI attacks demonstrate the breadth of vulnerabilities in AI systems. Several well-known examples exist. For example, the so-called 'panda-to-gibbon misclassification' refers to a case where an image of a panda was altered using imperceptible perturbations.[14] Although the modified image looked identical to humans, the neural network misclassified it with high confidence as a

[14] Goodfellow (2014).

gibbon (Figure 2.1). While this is a low-consequence example, other researchers placed small pieces of electrical tape on a traffic sign, causing an autonomous vehicle's pattern recognition system to misread a stop sign as a speed-limit sign.[15]

**Figure 2.1. Demonstration of an adversarial AI attack**




Note: a little seemingly random noise is added to an image, causing an AI object identification algorithm to misclassify it, while the change is imperceptible to a human. Source: Goodfellow (2014).

A more recent example is a vulnerability in Microsoft 365 Copilot that allowed unauthorised actors to execute malicious AI commands remotely. Exploitation of the vulnerability enabled attackers to disclose confidential data and execute commands on the victim's Microsoft 365 applications.[16] AI systems that provide public-facing services can be copied or misused by malicious actors. Researchers have shown that by interacting with these publicly available tools – such as online translation services – attackers can build replicas. These copies can then be used to study how the original system works and to craft inputs that intentionally cause it to produce incorrect or misleading results, potentially fuelling disinformation, misinformation or radicalisation.[17]

### 2.1.2. Depending on the type of attack and the AI system affected, Adversarial AI attacks could have significant consequences in the military domain that range from tactical disruption to broad escalatory risks

Much of the current discourse on military AI focuses on Autonomous Weapons Systems (AWSs), where concerns centre on the risk of unintended engagements. The United Nations Institute for Disarmament Research (UNIDIR) highlights misclassifications (whether arising from environmental conditions such as fog or dust, or from deliberate adversarial manipulation) as potentially having significant consequences – both for the efficacy of military operations and for compliance with obligations under international humanitarian law (IHL) – using the example of a military vehicle being mislabelled as civilian or vice versa in the course of targeting.[18]

[15] Evtimov (2018); and see Figure 2.3.

[16] Ravia (2025); Reddy (2025).

[17] For more examples, see the MITRE ATLAS™ case studies in deployed AI systems (As of 25 September 2026): https://atlas.mitre.org/studies

[18] Holland Michel (2021).

The following sections provide a structured overview of different categories of adversarial attacks and the range of AI systems they may affect, enabling a systematic assessment of potential implications.

## 2.2. Threat vectors

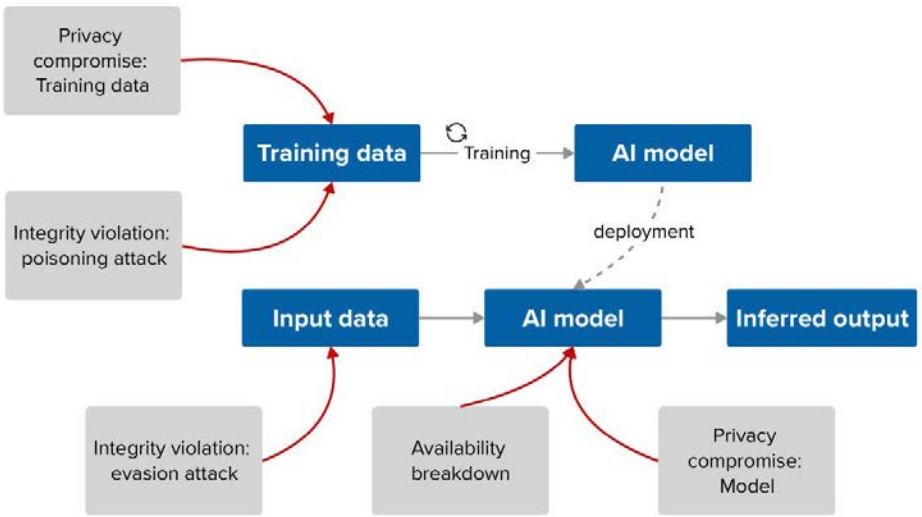
### 2.2.1. Research suggests there are three main categories of Adversarial AI threats

Drawing on academic literature, we distinguish three threat vectors, each linked to a different potential attack goal. A recent taxonomy developed by the US National Institute of Standards and Technology (NIST)[19] classifies the attacker's objectives according to three dimensions[20]:

1. **Availability breakdown attacks** prevent or limit timely and reliable access to an AI system's outputs. While the attack may be prepared during training or deployment, the effects are experienced when the AI system is deployed and in use.
2. **Integrity violation attacks** cause an AI system to perform poorly, with misclassifications that align with the attacker's goals.
3. **Privacy compromise attacks** aim to disclose confidential or proprietary data. This data can include training or operational data as well as the AI model's underlying weights or architecture.

Figure 2.2 shows an exemplar subset of these attacks relative to the AI model process, distinguishing training-stage and deployment-stage attacks. Training is the process of teaching an AI model to recognise patterns by exposing it to large datasets and adjusting its internal parameters to improve accuracy. Deployment is when the trained model is integrated into real-world systems to process live data and generate predictions or decisions. We discuss each threat vector in more detail in the following sections.

Figure 2.2. Type of adversarial AI attacks




Source: structure from Studer et al. (2021); threat vectors from Vassilev et al. (2025).

[19] Vassilev et al. (2025).

[20] This classification aligns with the classic cybersecurity triad of Confidentiality, Integrity and Availability (CIA) (Nieles, 2017).

Table 2.1. Goals and risks of adversarial AI attacks

| Attack type | Goal | Key impacts |
|---|---|---|
| Availability breakdown | Disrupt timely and reliable access to the AI system | Service downtime, delayed decisions |
| Integrity violation | Cause incorrect outputs or misclassifications | Operational failures, wrong decisions |
| Privacy compromise | Reveal confidential or proprietary information | Data breaches, IP theft, compliance violations |

Source: Vassilev et al. (2025).

### 2.2.2. Availability Breakdown attacks limit access to the AI system

Availability attacks aim to disrupt or degrade the normal functioning of an AI system, making it slow, unreliable or completely unusable. These attacks overwhelm the AI system or exploit weaknesses so that it cannot deliver timely or accurate results. Common examples include flooding the model with excessive queries to exhaust resources, introducing inputs that trigger costly computations and cause denial of service, or making the model produce nonsense output, rendering the system useless. One example is using 'sponge examples' that increase latency (the delay between providing input to an AI system and receiving the output) by up to 6,000 times.[21] Such a significant increase in latency could seriously disrupt military decision-support systems or autonomous weapon systems, slowing the Observe, Orient, Decide, Act (OODA)[22] loop and giving the adversary a significant advantage in high-tempo operations.

### 2.2.3. Integrity violation attacks cause the AI system to produce incorrect outputs

Integrity violation attacks are adversarial techniques that deliberately compromise the correctness of an AI system's outputs, causing it to produce inaccurate or misleading results. These attacks can occur at different stages of the machine learning lifecycle. During AI model training, attackers may launch poisoning attacks by injecting malicious or mislabelled data into the training set, leading the model to learn incorrect associations that persist after deployment (see Figure 2.3). At deployment, adversaries can also use so-called evasion attacks, applying physically or digitally perturbed inputs that exploit model vulnerabilities and trigger misclassifications (see Figure 2.3).

Integrity violation attacks could pose significant risks. For example, researchers have shown how poisoning attacks can manipulate military object-detection systems, causing them to misclassify military vehicles and thereby undermine situational awareness and operational decision making.[23] A joint study by the UK AI Security Institute (AISI), the Alan Turing Institute and Anthropic recently showed that these attacks are more practical than previously thought. They showed that introducing just 250 malicious documents (0.00016% of the full training dataset) can produce a vulnerability in a large language model (LLM), causing the model to produce nonsense output.[24]

[21] Shumailov (2021).

[22] The OODA loop consists of observing the situation (O), orienting by interpreting what those observations mean (O), deciding on a course of action (D) and acting on that decision (A), with results feeding back into the next cycle.

[23] Liezenga (2025).

[24] Souly (2025).

Figure 2.3. Mechanisms for two types of integrity violation attacks (evasion and poisoning)

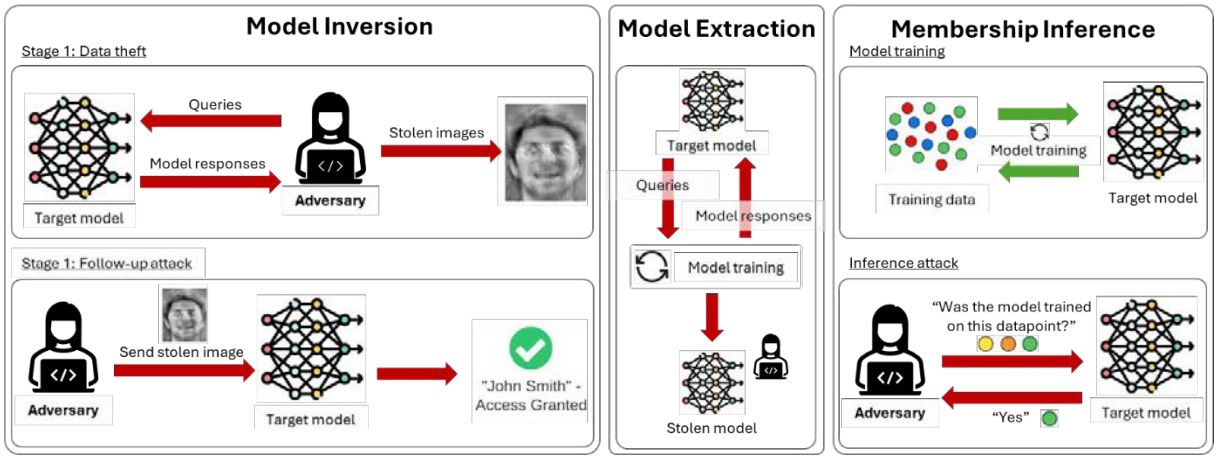



Source: adapted from Birch (2025).

### 2.2.4. Privacy compromise attacks disclose confidential or proprietary information from an AI system

Privacy compromise attacks occur when adversaries exploit an AI system to reveal confidential or proprietary information, posing risks to data protection and organisational security. These attacks can expose training data, such as personal, commercial or operationally sensitive data (membership inference or model inversion). Attackers may also target the underlying AI model itself, extracting its parameters and architecture, and enabling them to replicate or steal the system (model extraction). Generated outputs can inadvertently disclose sensitive information, as seen in cases where prompt injection manipulates LLMs to extract confidential information. Figure 2.4 shows how these privacy compromise attacks can work in practice.

Figure 2.4. Mechanisms for three types of privacy compromise attacks




Source: adapted from Birch (2025).

### 2.2.5. Adversarial AI attacks can also be distinguished by other dimensions: the attacker's capabilities, knowledge of the target, and the timing of the attack

Adversarial AI attacks can also be differentiated based on the characteristics of the attacker.[25] Understanding these factors provides a more nuanced view of the threat landscape. First, the NIST taxonomy distinguishes six capabilities an attacker can control and use for different types of attacks, as presented in Table 2.2. An adversarial AI attack may utilise one or multiple of these capabilities. Whether an attacker can exploit each capability depends on the attacker's access level to the AI system.

**Table 2.2. Attacker capabilities**

| Attack capability | Action |
|---|---|
| Training data control | Inserting or modifying training samples. |
| Model control | Directly changing model parameters. |
| Testing data control | Adding perturbations to testing data during deployment. |
| Source code control | Directly modifying the algorithm's source code, which can also happen through modifying third-party (open source) libraries used across AI systems and applications. |
| Query access | Submitting queries to the AI and receiving the output. This access is the most limited of the above, but can, for example, be used for privacy attacks or prompt injection. |
| Resource control | Resource control only applies to generative AI or AI systems that continue learning when deployed and involves modifying documents or webpages ingested by the generative AI during deployment, indirectly injecting adversarial information into the system. |

Source: Vassilev et al. (2025).

Second, the taxonomy distinguishes three different levels of knowledge that the attacker may have about the AI system, as presented in Table 2.3.

**Table 2.3. Attacker's level of knowledge about the system**

| Attack type | Knowledge about the system |
|---|---|
| White-box attacks | The attacker has full knowledge of the system, including training data, model architecture, and parameters. While rare, these attacks are used to assess a system's worst-case vulnerability. |
| Black-box attacks | The attacker has minimal to no knowledge about the system. |
| Grey-box attacks | Between white- and black-box, grey-box attacks cover a range of scenarios where the attacker knows the training data, the model and its parameters, or some combination of both. |

Source: Vassilev et al. (2025).

Third, timing matters. Adversarial AI attacks can occur during training or deployment, when the model generates predictions in real-world settings. During training, an attacker may influence the training data or labels, manipulate model parameters or interfere with the underlying code. These actions can lead to various forms of poisoning attacks that embed harmful behaviour into the model before deployment. Once the model is deployed, attackers can launch evasion attacks that cause integrity violations by tricking the model

[25] For generative AI, Vassilev et al. (2025) add a category of attacks: misuse enablement, which refers to the circumvention of restrictions by the system's owner. For example, Molas & Lopes (2024) show how far-right extremists circumvent ethical and operational constraints to generate harmful content that could facilitate radicalisation, recruitment, and the targeting of minority communities online. Although these attacks may affect a society's broader stability and resilience, this research does not consider them in depth.

into producing incorrect outputs. They may also carry out privacy attacks, attempting to infer sensitive information about the training data or the model itself.

## 2.3. Attack surface

### 2.3.1. As the AI technology stack becomes increasingly integrated into defence, the available attack surface for application of these techniques is evolving too, becoming simultaneously larger and more complex

Ranging from the tactical to the strategic, the current and future military applications of AI are diverse.[26] While these diverse applications offer opportunities to enhance effectiveness, efficiency, and lethality, they also expose new potential threat vectors for adversarial attacks.

Similarly, a complex, varied ecosystem of different actors is involved across the lifecycle of a military AI system, ranging from the frontier AI and defence tech firms to traditional defence industry, ministries of defence, militaries themselves and related institutions (e.g. wider government, intelligence and security agencies, or operators of critical national infrastructure). Some systems will be highly connected, while others are air-gapped; some will involve significant commercial actors, while others will be controlled, maintained and updated solely by military personnel. Defence organisations must also contend with difficult questions about tech sovereignty and interoperability when deciding how, when and with whom to integrate AI into military operations.[27] As the adoption of digital technology within defence proceeds unevenly, the latest AI models may also coexist for many years alongside legacy hardware and software. Rollout may also have to contend with outdated policy, data management, procurement and AI test or assurance systems and a deficit of digital skills. All these ecosystem features create seams, dependencies and vulnerabilities that attackers can exploit.[28]

### 2.3.2. Our study narrows the focus to those elements of the attack surface where adversarial AI attacks could have most significant effects on strategic stability

Against this backdrop, this study focuses on AI systems whose disruption through adversarial AI attacks could most affect strategic stability, either because of their criticality in civilian life or military missions, or because of the sensitivity of the data an attack could expose. For instance, while failures in self-driving vehicles would have serious consequences and could be weaponised for terror attacks, they are unlikely to reach the strategic level, unlike disruptions in targeting critical military systems.

The resulting list was informed by the discussions during our expert workshop (and can be found in Annex A.3), and subsequently consolidated into the five categories outlined below and detailed in the following chapter:

[26] Black et al. (2024).

[27] Huismans et al. (2025).

[28] Lucas et al. (2025).

- (L)AWSs
- Military decision-support systems
- Military back-end functions
- NC3
- CNI (civil or military).

While this report focuses on the implications of individual failures, it also highlights the risk of vulnerabilities in common technologies shared across AI systems. In conventional cybersecurity, a far-reaching example is a vulnerability found in Log4j, an open-source library widely used by apps and services. This vulnerability enabled attackers to break into systems, extract data (including passwords and logins) and infect networks.[29] AI systems often depend on shared technology or principles. For example, civilian object detection often depends on You Only Look Once (YOLO) models.[30] A shared vulnerability could enable a shared adversarial AI attack, with more far-reaching implications than the failure of an individual system.

Additionally, adversarial AI could affect AI adoption, proliferation, safety and security concerns more widely. These impacts would shape strategic competition and therefore have indirect, second-order effects for defence and security, e.g. around disinformation, the shifting determinants of national competitiveness, the rollout of agentic AI or the possible advent of Artificial General Intelligence (AGI). Such risks are discussed in the following chapter's final subsection.

[29] National Cyber Security Centre (2021).

[30] Sapkota (2025).

# 3. Manifestations of Adversarial AI

This chapter explores the implications of different illustrative adversarial attack scenarios. Each section describes a different attack surface and outlines a scenario showing how different types of adversarial AI attacks may affect the AI system and lead to adverse outcomes. These scenarios derive from a workshop convened with AI governance, adversarial AI, defence, and security experts and stakeholders. While not intended to be comprehensive, they illustrate different pathways through which adversarial AI threats could impact strategic stability, as a basis for further analysis in subsequent chapters.

### Box 6. Summary of findings: Chapter 3

**To illustrate how implications vary by attack type and target system, we present five illustrative scenarios.**

These scenarios comprise intersections of an adversarial AI attack type (threat vector) and an AI system type that could be affected (attack surface). The intersections, selected to present a diverse set of scenarios and to illustrate the breadth of implications for strategic stability, and informed by the expert workshop, are:

- **Attacks on (lethal) autonomous weapon systems** (Scenario 1: Integrity Violation)
- **Attacks on military decision support systems** (Scenario 2: Integrity Violation)
- **Attacks on military back-office functions** (Scenario 3: Integrity Violation, Privacy Compromise)
- **Attacks on NC3** (Scenario 4: Availability Breakdown, Integrity Violation)
- **Attacks on CNI** (Scenario 5: Availability Breakdown, Integrity Violation).

Each subsection below addresses one of five main types of AI systems, whose disruption is likely to impact strategic stability. Each section follows a similar structure. First, we lay out the attack surface and how AI is integrated; second, we explore the possible implications of an adversarial AI attack through an illustrative scenario. Each scenario addresses the immediate consequences of the adversarial AI attack, the factors that determine their severity, the plausible worst-case outcome and the implications for strategic stability.

To assess the implications for strategic stability, we draw on relevant frameworks such as crisis stability, arms-race dynamics and broader geopolitical risks, as well as deterrence concepts including imposing costs, denying gains, ensuring a secure second strike, maintaining credibility, attribution challenges and the potential for miscalculation or mistrust. To aid the reader, Box 7 presents a glossary of strategic stability and deterrence concepts to help situate the discussion of each of the illustrative scenarios – and the more detailed analysis of the implications of adversarial AI for strategic stability that follows in Chapter 4.

### Box 7. Glossary of relevant deterrence concepts

- **Denying gains:** making it difficult or impossible for an adversary to achieve meaningful benefits from an attack, thereby discouraging them from attempting it.
- **Imposing costs:** threatening or ensuring meaningful punishment (military, economic or political) if an adversary attacks, making the expected costs outweigh potential gains.
- **Offence/defence balance:** the relationship between how easy it is to attack versus how easy it is to defend; when defence is stronger, deterrence is generally more stable.
- **Potential for miscalculation or mistrust:** the risk that states misunderstand each other's intentions or capabilities, which can lead to unintended escalation or conflict despite deterrence efforts.
- **Secure second strike:** the guaranteed ability to retaliate after absorbing a first attack, ensuring an adversary cannot eliminate the response capability.
- **Credibility:** the degree to which an adversary believes a state has the capability and the will to follow through on threats or commitments.
- **Attribution:** the ability to accurately identify who carried out an attack.

Source: Black (2024); Mazarr (2018).

## 3.1. Autonomous weapon systems

Autonomous weapons systems (AWSs) are a category of capability that incorporates autonomy into some or all functions – for example, navigation and target selection.[31] Current capabilities are not fully autonomous and require a human operator, either 'in-the-loop' or 'on-the-loop', to issue commands or override actions.[32] While the term AWS encompasses many varied weapons systems (elements of autonomy are present/could be implemented in systems across the force, such as air defence systems or smart munitions), the most prominent examples are uncrewed aerial, maritime or land vehicles (UxVs) such as the 'loitering' systems in use in Ukraine. While human agency remains a necessity in AWS for now, fully autonomous AWSs are possible in the future – systems that, upon activation, can select targets, manoeuvre and engage with lethal force without human intervention.

### 3.1.1. AWSs represent the most prominent area of debate in relation to military AI generally, and adversarial AI attacks specifically

The growing integration of AI into robotic and uncrewed systems (UxS) operating on the frontline represents an emerging vector for adversarial attacks in the military domain. Introducing AI support into uncrewed Intelligence, Reconnaissance and Surveillance (ISR) or logistics and lift systems is desirable for militaries worldwide for several reasons. Primarily, AI-enabled capabilities offer speed, efficiency and resilience while reducing risk to human personnel.[33] For example, an AI-enabled air defence system can track massed, fast-moving emerging threats such as drone swarms more effectively and efficiently. Similarly, AI-enabled UxS can continue operating even in the face of denied communications back to a remotely

[31] Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons System (2023).

[32] Mosqueira-Rey (2023); Scharre (2018).

[33] Development, Concepts and Doctrine Centre (2023).

located human pilot or commander, making them valuable tools for operations in a highly contested electromagnetic environment.[34] However, as autonomy levels continue to grow and reliance on AI increases, the potential implications of adversarial attacks on these systems aggravate.

Perhaps the most high-profile debate in this space concerns the potential for Lethal Autonomous Weapons Systems (LAWSs). Armed systems in the contemporary battlespace (e.g. Ukraine, the Middle East) are becoming increasingly independent, such as 'loitering' UxS.[35] These systems can be deployed to an area and wait for a target to move within range, autonomously monitoring the environment. Once a target has been identified, the UxV contacts a human operator, who can either approve a kinetic strike (human-in-the-loop) or, where greater levels of machine agency may exist, override a system's decision to launch a strike (human-on-the-loop).

Fully autonomous kinetic systems (so-called 'killer robots') without a human in- or on-the-loop are unlikely to be rolled out at scale in the near term, due to both ethical considerations and the practicalities of implementation. For its part, the UK has stated that it does not possess, nor intends to develop, full AWSs; instead, it will retain Contextually Appropriate Human Involvement (CAHI).[36] Therefore, a human is likely to remain at least 'on-the-loop' for kinetic operations, providing authorisation for the target prosecution phase (although the same cannot necessarily be said for adversaries).[37] CAHI should, in theory, reduce the impact of an attack. For example, a human operator in a human-on-the-loop AI system may detect integrity-violation attacks, limiting the likelihood of immediate harm. However, automation bias (the human tendency to favour suggestions from automated systems), limited time for deliberation, and a possible lack of system understanding may reduce the rigour of human oversight,[38] exacerbating possible implications of adversarial attacks despite CAHI.

Beyond the target prosecution and firing decision, adversarial AI attacks could also target the non-kinetic supporting functions of AWSs, such as autonomous navigation for UxS, object detection and target identification. Disruptions of the navigation system, for example, could cause a UxS to crash or fail to reach its target. Malfunctioning object detection or target identification could cause the UxS to miss the opportunity window to engage or fail to detect a hostile incursion. Availability breakdown attacks could break down communications and limit the operator's ability to control the system. Ultimately, the loss of CAHI in AWSs would limit resilience to adversarial attacks, generating threats to strategic stability as human agency and thus control over volatile situations decreases.

A further consideration is whether and how any AWS targeted by adversarial AI attacks are integrated into a wider system of systems (e.g. a so-called 'kill web'). If AWSs are highly networked, and therefore connected to and cued by crewed ISR platforms or human commanders, an attack on the AI may have more limited effects, as there will still be opportunities for human intervention. Conversely, AWSs operating without

[34] Robinson (2023).

[35] Aviram (2025).

[36] UK Ministry of Defence (2022a).

[37] Perrin (2025).

[38] Watts (2023).

direct tasking by higher echelons or connections to a wider network, for example due to a contested electromagnetic environment, may pose greater risks if their onboard AI algorithms are compromised.

### 3.1.2. Illustrative Scenario 1: Integrity violation attacks on AWSs lead to unintended escalatory action and collateral damage

The expert workshop highlighted integrity violations of AWSs as the most immediately concerning threat vector at the tactical level. Because of the system's kinetic nature, the immediate consequences of misidentifying a target could be severe. While an attack on an autonomous ISR asset, for example, might compromise mission objectives, an integrity violation of the targeting system of an AWS or strike UxS could lead to collateral damage, unintended escalatory action, or even fratricide (firing on allied systems/units).

Workshop participants assessed that a primary factor determining the severity of the attack would be the AWS's level of autonomy, especially regarding targeting. Human oversight (on-the-loop, if not in-the-loop) could help to mitigate incorrect classification directly and prevent failure – for example, with the human recognising a selected target to be, in fact, civilian rather than military and overriding the strike. However, human-out-of-the-loop systems lack this additional, if imperfect, safety net. Without a human operator as a failsafe, a compromised AWS could carry out unauthorised escalatory action (for example, accidentally prosecuting a civilian or friendly target), with obvious destabilising effects in a crisis.

If the integrity attack affects only one individual system, the impact would likely be limited to the tactical level. For example, the failure of a single AWS due to an evasion attack during deployment is unlikely to have a significant impact beyond the immediate mission. However, if the attack occurs earlier in the machine learning lifecycle, e.g. a poisoning attack during the training stage, it could compromise multiple systems that use the AI model. As the scale of the attack increases, so too do the related impacts. As failures of multiple systems stack up, the chances of an accidental escalatory action (of the same kind as those discussed above) increase, and it becomes obvious that a proportion of a state's conventional military capability has suddenly become unreliable. Deterrence is based on the belief that it would not be in an actor's favour to initiate conflict with a state/alliance. If an attacker can compromise a set of AI-enabled capabilities rather than just individual systems, they could therefore damage the broader credibility of the targeted nation's forces and their perceived ability to respond to a crisis with their uncrewed assets – compromising strategic stability by undermining the basic logic of conventional deterrence.

## 3.2. Military Decision Support Systems

Decision Support Systems (DSSs) are tools that help decision makers evaluate options and provide the information needed to make informed decisions in complex situations.[39] In the military context, this encompasses a broad range of systems to support tactical, operational and strategic decision makers – everything from planning an individual mission to long-term processes such as force (personnel and weapons) management. While DSSs traditionally rely on automation (following pre-programmed actions), integrating AI introduces autonomy and learning, making them more adaptable and responsive. For

[39] Nadibaidze et al. (2024).

example, an AI-enabled DSS might predict likely enemy force movements, whereas a traditional system might simply plot formation locations on a map based on inputted data.

### 3.2.1. AI is likely to facilitate decision support across all activities, from data collection and sense-making to planning and execution

AI integration into decision support systems is increasingly seen as an operational imperative by militaries as time-space dynamics contract across the battlespace. As the balance shifts away from slow, deliberate planning cycles like the 72-hour air mission planning process toward on-the-fly, dynamic decision making, low-latency AI-enabled support systems will likely become critical to warfighting performance.[40]

AI is likely to facilitate decision making across all activities, from data collection and analysis/sense-making to planning and execution, with possible applications across many aspects of ISR[41] and C2.[42] For example, AI is being explored as a tool to transform the Task, Collect, Process, Exploit, Disseminate (TCPED) cycle[43] – particularly around the Process step, which involves the fusion, processing and prioritisation of huge amounts of data.[44] Advanced future decision support systems could generate Course of Action (COA)[45] options, run predictive analytics and simulations to adjust those COAs and distribute the corresponding orders, involving AI (at least to some extent) in every step of the OODA loop.[46]

As a result, these steps are also becoming vulnerable to adversarial attacks, and successful attacks could have widespread implications. Whereas an attack against an autonomous weapons system might lead to localised failure of an individual capability, decision support systems directly influence human tactical, operational, and strategic decision making.[47] Decision-support systems present an attack surface that could lead not only to the failure of a given system but also to cascading second- and third-order effects.

### 3.2.2. Illustrative Scenario 2: Integrity violation attacks on AI-enabled decision support systems subtly distort human decision making

AI-enabled decision support systems are equally, if not more, vulnerable to integrity violations as AWSs due to the significant volumes of data processed through them – most of which could be exposed to manipulation through adversarial attacks. The consequences of an attack on decision support systems are,

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[40] Lingel et al. (2020).

[41] ISR is the coordinated acquisition, processing and provision of accurate, real-time information to support military operations and decision making. The individual elements of ISR are Surveillance (monitoring of a target), Reconnaissance (gathering information to answer a specific military question), and Intelligence (the product of Surveillance and Reconnaissance). Source: North Atlantic Treaty Organization (2025).

[42] King (2024).

[43] TCPED are the five subordinate processes of the ISR process, consisting of receiving and managing the Task (T), collecting information (C), processing raw data into a usable format (P), deriving value from the information (E) and disseminating the data, information and intelligence to the relevant parties (D). UK Ministry of Defence (2022).

[44] Development, Concepts and Doctrine Centre (2023).

[45] COAs are the different realistic options a commander can choose from to achieve the mission under given conditions.

[46] Schubert et al. (2018).

[47] RAND Europe expert workshop (December 2026).

perhaps, more subtle, but potentially more concerning. AI-enabled decision support systems are a form of human-machine teaming – fundamentally designed to influence how the human commander thinks.[48] A high-profile example is the US Project Maven Smart System, which provides military operation targeting support through data integration and visualisation.[49] Successful adversarial attacks against such systems would affect not just the system itself, but also how users interact and make decisions, and consequently their trust in technology and their situational understanding.

The severity of such an attack would partially depend on where AI-enabled decision support systems are deployed in the future. The core risk of this threat vector is the distortion of human decision making (as opposed to the degradation of the system in itself), and so severity would naturally increase as the decisions a system can influence become more critical. Decision-support systems are already being deployed at the tactical and operational level. However, if strategic-level actors were to adopt similar tools, it would expose a threat vector for adversarial attacks that could realistically impact decision making at the highest level – and thus strategic stability. In the same way that current decision support systems might influence targeting decisions, future decision support systems deployed at the strategic level could influence decisions over force structure, posture or procurement (for example). If an adversary were able to compromise these systems through an integrity violation, they could both have an impact on capability – perhaps changing the offence-defence balance – and influence the behaviour of allied strategic actors when it comes to crisis management or escalation management during a live conflict. This situation would create greater unpredictability, generating opportunities for miscalculation at the highest level.

Regarding decision-support systems used for targeting, the expert workshop found that adversarial attacks on ISR and Sense, Decide, Effect (SDE) capabilities could have operational implications. To offset a lack of operating mass, the UK instead focuses on precision through advanced ISR and SDE to achieve maximum effect with limited forces. AI integration into these systems is underway, most notably through the flagship Project ASGARD initiative and an AI-enabled Digital Targeting Web (DTW) designed to enhance the lethality of the UK's forces.[50] However, the shift toward autonomy as a solution to a lack of mass simultaneously generates greater dependencies. An adversarial attack against systems like ASGARD could cripple C2 and ISR capability, and thus the ability of the affected forces to operate – and deter - effectively.

The strategic implications in this scenario would depend largely on the reversionary modes available in the event of a disruption or malfunction, and on the capability to contain the attack and restore function.

## 3.3. Military back-end functions

Integrating AI across defence more broadly encompasses a wide range of potential systems, including the introduction of AI-enabled systems across back-end support functions, not just among Frontline Commands. This possibility could include everything from resource management (e.g. logistics and

[48] Nadibaidze et al. (2024).

[49] Mohsin (2024).

[50] UK Ministry of Defence & Healy (2024).

procurement) to human processes (e.g. recruitment and training). Ultimately, most (if not all) supporting processes behind a successful defence enterprise could benefit from AI integration for purposes such as optimising resource allocation or real-time monitoring of the performance of large-scale and long-term acquisition programmes.

### 3.3.1. AI integration in the back-end functions of defence organisations could range from logistics, procurement or training optimisation and HR functions to Research and Development (R&D)

AI integration is also occurring across the defence enterprise more broadly, not just at the 'tip of the spear.' Indeed, many of the tasks AI performs well in (such as data processing) arguably offer an equal, if not greater, advantage to defence in these longer-term, back-end functions than AI integration on the frontline.[51] Supplemented by our workshop findings, the UK's Defence AI Playbook identifies a diverse range of potential AI applications in this space, including logistics optimisation, procurement systems, training optimisation, HR functions and R&D, alongside the more obvious operational applications.[52]

As a result, the broader defence enterprise (including the military, government and industry) presents an expansive attack surface that extends far beyond direct use of AI systems by frontline commands. Concerningly, these threat vectors are perhaps less obvious than headline implementations in weapons systems and, as such, may not be as thoroughly scrutinised or well protected.[53] This is particularly true for functions involving the private sector and/or academia (such as procurement or R&D), which could enable adversarial attacks not just on defence directly, but across a wider supply chain and network.[54] Attacks do not have to target defence directly to have a significant impact, exposing an even wider attack surface.

### 3.3.2. Illustrative Scenario 3: Integrity violation or availability breakdown attacks on back-end functions unfold slowly but have wider, enterprise-level impacts

Although AI deployment is expected across a multitude of functions, outputs from the expert workshop focused on three broad areas where adversarial attacks could have an outsized impact: procurement, logistics and Human Resources (HR). Crucially, participants considered back-office processes 'slower' than their frontline counterparts, meaning disruptions or manipulation through adversarial AI threats can propagate deep into the organisation. Attacks on such systems degrade not just a given AI system, but the processes that underpin the defence enterprise's effective functioning, with cascading effects on its efficiency.

Procurement is typically a slow process even when compared to other back-end defence functions, taking years if not decades for some capabilities to reach the user. This characteristic creates a certain level of resilience, and short-term adversarial attacks such as access limitation/denial are likely to have a negligible impact on such long-term processes, since any single instance would be unlikely to shift timelines perceptibly. In contrast, privacy attacks could cause AI-enabled procurement systems to leak sensitive information about future capability or industry Intellectual Property (IP). If an adversary could successfully

[51] UK Ministry of Defence (2022).

[52] UK Ministry of Defence (2024a).

[53] Black et al. (2024).

[54] RAND Europe Expert Workshop (December 2026).

orchestrate such an attack against AI tools involved across the lifecycle for a critical acquisition, e.g. submarines, combat aircraft, or complex warships, it could compromise the effectiveness of the capabilities that are supposed to underpin conventional and possibly nuclear deterrence for the foreseeable future.

Further, procuring AI systems is itself a related but distinct privacy-compromise attack vector prioritised in the expert workshop. Specifically, if an adversary could steal AI models and their weights, it could prevent effective use of the model by developing countermeasures or negate potential advantage by replicating the system itself or proliferating it to allies and proxies. While this would have tactical- and operational-level impacts on the force options available to commanders, implications could also bleed into the strategic level, depending on the AI system's criticality.

Participants also identified AI-enabled military or commercial logistics systems as a potential target. The ability to react quickly to dynamic threats is a deterrent in itself, and capable logistics systems are a key supporting element to power projection. A successful adversarial attack on the supporting systems involved in these deployments could prevent or slow the targeted nation from engaging in operations beyond its immediate homeland, reducing its credibility among both allies and adversaries.

Finally, HR is a major area of defence that is adopting AI systems, particularly in recruitment and personnel management processes. While access attacks on these systems would become obvious relatively quickly, and a reversionary mode (the human HR process) should remain a viable option, integrity violation and privacy compromise attacks pose a much greater risk in terms of the second-order impacts they could generate around defence's people. For example, an integrity violation of an AI-enabled recruitment system could lead to recruiting personnel to frontline services who should have been ineligible due to medical conditions, or a privacy compromise of sensitive personal data could render defence personnel vulnerable to blackmail. Whatever form these second-order impacts take, they may not be immediately obvious – and as the affected proportion of defence grows over time, these implications could evolve into strategic, system-level vulnerabilities that compromise defence as a whole.

## 3.4. Nuclear command, control, and communications

Nuclear command, control and communications (NC3) refers to the 'facilities, equipment, communications, procedures, and personnel' that enable direction of a state's nuclear forces.[55] It encompasses a broad range of activities associated with the nuclear enterprise, including managing nuclear forces, planning for the use of nuclear weapons, monitoring information on friendly and adversarial nuclear forces, making decisions on nuclear use and implementing these decisions. While the extent to which AI is already integrated into NC3 (if at all) is unclear, core components of NC3 systems that could potentially integrate AI in the future include sensors (e.g. early warning satellites such as the US space-based infrared system satellite constellation for detection of ballistic missiles), data-transfer systems (e.g. radio frequency systems or SATCOM) and situational awareness tools (e.g. the US Integrated Tactical Warning and Attack Assessment System, which provides information on missile, space and air attacks).

[55] Deptula et al. (2019).

### 3.4.1. While full autonomy in NC3 is improbable, partial AI adoption is likely

Although this study focuses on non-nuclear strategic stability, NC3 remains a crucial element to consider. While the consensus is that removing human input and/or oversight of NC3 entirely is improbable, it is likely that AI will, to some extent, be implemented in this sphere in the future.[56]

NC3 is expansive, including everything from early warning systems (including assets such as satellites and radars) to communications networks connecting decision makers to operators, as well as the nuclear weapons launch platforms themselves.[57] Integrating AI into some of these systems, e.g. to speed up data processing to enable faster differentiation between conventional and nuclear warheads, could become attractive to nuclear-armed states as decision-making windows continue contracting.[58] As such, certain parts of the overall NC3 system of systems could become exposed to adversarial attacks, even if not to the same extent as conventional systems given the additional security measures in place (e.g. air-gapping, high encryption levels and reduced use of non-sovereign or commercial tech).

NC3's entanglement with conventional NC3 poses a potentially thorny issue. Because C2 systems in the United States, Russia and China support both conventional and nuclear forces, any attack on them could cause uncertainty about the attacker's intent and increase the risk of inadvertent escalation.[59] Where networks overlap, the attack surface associated with nuclear targets expands. In such cases, ostensibly conventional attack surfaces have implications in the nuclear domain, with potential consequences for strategic stability due to the risk of miscalculation or inadvertent escalation.[60]

### 3.4.2. Illustrative Scenario 4: Integrity violation or availability breakdown attacks on any AI-enabled NC3 could risk hyper-caution and (inadvertent) escalation

While difficult to achieve, successful infiltration of any AI system that limits or denies access, or violates integrity, could have catastrophic consequences once a state's nuclear arsenal is at risk. Any attack that impacted the ability of a state to communicate with its nuclear forces could conceivably be perceived as an attempt to deny that state's assured second strike capability. By its very nature, this undermines strategic stability, potentially sparking a nuclear escalation spiral following the logic of crisis instability – that when assured second strike is under threat, nuclear decision makers are pushed toward 'use-it-or-lose-it' choices with the highest of stakes.

Integrity violation attacks specifically introduce another layer of uncertainty to this equation, increasing the chance of miscalculation. The potential for these attacks to go unnoticed for longer periods of time – as opposed to access limitation, which would be immediately obvious – introduces the threat of unreliability of NC3 and the intelligence feeds shaping decision making in a nuclear crisis. Data poisoning or model manipulation could feed false situational awareness, suggesting non-existent threats or hiding real ones. As

[56] Mennesson (2025); RAND Europe Interviews (January 2026); Saltini (2025b).

[57] Saltini (2023a).

[58] Hoell & Mishra (2023); Saltini (2025a).

[59] Acton (2018); Talmadge (2017).

[60] Acton (2020); van Hooft (2026).

a result, nuclear decision makers could become overly cautious (risking the breakdown of deterrent credibility), or, in a worst-case scenario, overly pre-emptive (risking major escalation).

Entangled conventional and nuclear C3 presents the most exposed threat vector in this space. An adversarial attack may not be able to distinguish between the nuclear and conventional functions of a dual-capable AI-enabled C3 system, adding yet another layer of complexity to the escalation calculation. If an attack occurred against a primarily conventional system that could incidentally affect NC3 – a military communications network, for example – it could be perceived as an attempt to cripple assured second strike. Ideally, any attack surface that could affect a state's perception of the security of its nuclear arsenal should be avoided to maintain strategic stability, but this is becoming more difficult as AI integrates into potential dual-capable C3 functions.

## 3.5. Critical national infrastructure

The use of AI in critical national infrastructure (CNI) could be, and arguably already is, widespread. While common tasks include cybersecurity (e.g. network monitoring and anomaly detection) and monitoring large-scale systems, sector-specific uses also vary (i.e. AI use in the healthcare sector will differ from its use in finance).[61] The impact of an adversarial attack on any given system could vary significantly based on these differences.

### 3.5.1. CNI presents a broad, exposed attack surface

Beyond the military, the safety and resilience of CNI is crucial to defence and security. These systems support the government, economy and society as a whole. Consequently, adversarial attacks against them have significant implications for strategic stability. AI technologies are already being integrated into CNI, with use cases including modelling and simulation, threat/event detection and analysis tools such as forecasting systems.[62] A 2024 report by the US Center for Security and Emerging Technology (CSET) identifies the financial, energy and water sectors as the most AI-mature.[63] For example, smart grid systems in the energy sector are incorporating AI for balancing and optimisation processes, as well as fault detection and classification.[64] AI integration is also increasing across other sectors, including healthcare, transport, agriculture, cybersecurity and communications. As adoption picks up pace, the potential attack surface and associated vulnerabilities or interdependencies expand exponentially.

Civilian CNI also presents a far more exposed attack surface than defence-specific counterparts. For example, communications networks are likely to generate significant vulnerabilities in the near future. As 5G communications networks expand, AI will (and arguably already has) become a core asset used for network slicing, automatic modulation recognition and power allocation.[65] 5G communications are also a crucial enabler for other sectors (including transport, energy and defence), multiplying the potential impact

[61] Crichton et al. (2024).

[62] Crichton et al. (2024)

[63] Crichton et al. (2024).

[64] Ardito et al. (2022).

[65] Apruzzese et al. (2019).

of an attack. At the same time, these networks are particularly vulnerable to adversarial attacks because of their open nature. Because access requires minimal verification, the barrier for attackers to access and manipulate the entire ecosystem is remarkably low.[66]

### 3.5.2. Illustrative Scenario 5: Integrity violation or availability breakdown attacks on critical infrastructure disrupt both military and civilian functions and will to fight

AI integration into CNI creates a readily exploitable threat vector that could have significant implications not just for defence, but for civilian life too. AI system integration across sectors creates potential vulnerabilities across society, threatening national resilience. This vulnerability, in turn, could threaten strategic stability. As per NATO's definition, resilience is 'the individual and collective capacity to prepare for, resist, respond to and quickly recover from shocks and disruption', and it is a central pillar of a nation's credible deterrence and defence.[67] Compromising national resilience undermines strategic stability by eroding civil preparedness and preparing the ground for further escalation. For example, adversarial attacks on critical sectors such as energy and water during a conflict could erode a nation's will to fight (the disposition to fight, and keep fighting, until victory) by limiting access to crucial resources.[68]

CNI is not a new target for malicious actors, but adversarial AI heightens the threat posed. CNI such as connectivity/communications networks and energy systems are a battleground for pre-conflict grey zone activity. Adversarial attacks, however, offer malicious actors new opportunities to attack CNI without a physical footprint, and potentially without anyone knowing an attack even took place. It provides a direct channel to threaten broader national interest while remaining ambiguous and unattributable.

Space-based services demonstrate how AI integration could expose both society and defence to novel adversarial attacks. They have become essential to a diverse range of daily functions, and systems exploit AI for tasks such as bandwidth allocation, satellite image interpretation, and anomaly detection.[69] This implies that an adversarial attack on AI-enabled space infrastructure and services could cause widespread disruption across the economy and the networked modern military.[70] For defence, attacks on SATCOM would go far beyond disruption of tactical and operational activity. It is an inherently strategic network that links the disparate nodes of the enterprise; its disruption could compromise the ability to organise a coherent response to a crisis. Adversarial attacks against such a target could simultaneously damage the credibility of conventional deterrence and erode national resilience.

## 3.6. Other impacts of adversarial AI on strategic competition

The previous subsections discussed the five areas of the attack surface most directly relevant to defence that our analysis and expert workshop suggest as particular concerns in relation to adversarial AI. However, adversarial AI could also affect AI adoption, proliferation, safety and security concerns more widely. While

[66] Apruzzese et al. (2019).

[67] North Atlantic Treaty Organization (2024).

[68] McNerney et al. (2018).

[69] Kumar & Sankar (2025).

[70] Department for Science, Innovation and Technology (2025).

not the focus of this study, these broader impacts of attacks on AI could nonetheless shape strategic competition and therefore have indirect, second-order effects for defence and security.

Four such indirect impact pathways are especially prominent in the literature:

- (Adversarial) AI impacts on disinformation
- (Adversarial) AI impacts on the determinants of national competitiveness
- Risks around Agentic AI
- Risks around Artificial General Intelligence (AGI) and loss of control.

The final subsections briefly address each in turn, before Chapter 4 discusses the overarching implications and key variables affecting the level of impact from a strategic stability perspective.

### 3.6.1. Adversarial AI could exacerbate the challenge of AI-enabled generation and dissemination of disinformation, affecting decision making and will-to-fight

While information manipulation is a core risk associated with AI more generally, adversarial AI capability could accentuate it. Generative AI (including LLMs) is being deployed throughout society for diverse tasks and can spread information faster, in more tailored ways, and in greater volume than any previous channels. Adversarial attacks – integrity violations in particular – could compromise this data or data used by non-GenAI systems, potentially with strategic implications.

Poisoning the data of an AI model could produce misinformation about politics and voting, for example, enabling belligerent fringe actors to gain political power and destabilising international relations.[71] Targeted disinformation campaigns using generative AI are already a concern, giving adversaries an indirect ability to influence both domestic and international audiences, e.g. to undermine alliance cohesion, sow doubt about hostile grey-zone operations or undermine public support for a military operation. Adversarial attacks could introduce a new, subtler element to this influence, allowing adversaries to covertly poison the data of nominally trustworthy models/services. For example, as many online newspapers are introducing AI assistants to help users find relevant articles, an integrity violation could feed users targeted, politically charged content. This capability gives adversaries a way to manipulate collective or individual perceptions and behaviour, potentially driving unintended escalation in a crisis or undermining will-to-fight.

### 3.6.2. AI could reshape the determinants of national competitiveness – militarily, economically and politically – meaning that adversarial AI threats could have macro-level effects on global competition for strategic advantage

Prior RAND research has shown that national competitiveness at the strategic level is not determined only by raw potential (e.g. population size, territory and GDP) but also by states' propensity to mobilise and translate this latent societal potential into useful outputs that support national goals.[72] Given its general-purpose nature, the rollout of AI technology has the potential to influence many of the determinants of

[71] Matza (2024).

[72] Black et al. (2023).

national competitiveness,[73] whether militarily, economically, politically or societally – for example, by unlocking sizeable new cognitive resources, enabling creation of new military capabilities (e.g. wonder weapons) or commercial products and services or providing leverage over the global AI tech stack.[74]

Strategic advantage in AI can be achieved not just by developing advanced systems but by rolling them out quickly, regulating them well and building the capacity to turn technological potential (e.g. frontier AI models) into real advantages (e.g. applications with meaningful operational benefits). This means investing in infrastructure, attracting and training talent and encouraging fast but responsible adoption across sectors.[75] At the same time, states can try to slow rivals' progress by limiting their access to AI talent and technology or by using adversarial AI attacks to extract from or disrupt an adversary's AI systems. In this sense, adversarial AI capabilities are a key component of strategic competition over and through AI, with potentially destabilising effects on relations between major powers if they perceive themselves to be in an AI arms race.[76]

### 3.6.3. Agentic AI systems pose unique, increased risks due to their autonomy and adaptability

Agentic AI further complicates the safeguarding and governance of military and critical AI systems. AI agents autonomously work toward defined goals, adapt to their environment and can potentially interact with the physical environment to achieve them. They plan, decide and act autonomously over multiple steps, with access to tools, databases, and real systems. Examples of civilian use of AI agents include making bookings and executing trades.[77] Their autonomy and adaptability pose unique adversarial AI risks, including cascading failures and increasingly difficult-to-anticipate emergent behaviours. Meanwhile, their interaction with the external environment opens a wide attack surface. An adversarial AI attack unique to AI agents is so-called *agent hijacking*, where an attacker inserts malicious instructions into data that one or more AI agents may process, causing unintended harmful actions.[78]

### 3.6.4. Adversarial AI threats pose a wider safety concern in relation to any future development, deployment or containment of AGI

Finally, while this report focuses on so-called Narrow AI – algorithms and systems that excel in a specific task – AGI and Artificial Super-Intelligence (ASI) are AI systems capable of performing intellectual tasks at or above human level. AGI/ASI are theoretical for now, but if realised, could rapidly transform the economy, military operations and the cyber domain, and cause systemic shifts in the balance of power, potentially even posing existential risks to humanity.[79]

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[73] Mazarr (2022).

[74] Mitre & Predd (2025).

[75] Kahl (2025).

[76] Mazarr (2026).

[77] IBM (2026)

[78] National Institute of Standards and Technology (2025).

[79] Black (2026); Mitre & Predd (2025); Negele et al. (2025).

Prior RAND research for the UK AI Security Institute and others has found adversarial AI attacks to be one of the most concerning threat vectors that could lead to a loss of control over an AGI.[80] Even if the many difficult and currently unsolved containment and alignment issues associated with AI safety in the context of AGI were resolved at both the technical and policy levels, privacy compromise attacks or model extractions could cause AGI systems to fall into hostile state or non-state actors' hands; these may have less robust oversight mechanisms, causing a loss of control over the system.[81] Similarly, there may be perceived first-strike incentives to try and use adversarial AI attacks to slow or disable an adversary from achieving AGI in the first place. However, success is not guaranteed, and this may spark a broader conflict.[82] Another possible pathway to strategic instability is an adversary manipulating rather than stealing or disabling a high-performing/AGI model, potentially resulting in model misuse – a well-known example of which would be jailbreaking a model to obtain (bio)weapon blueprints with strategic-level effects.[83]

[80] Boudreaux et al. (2025).

[81] van Soest (2025).

[82] Rehman, Mueller & Mazarr (2025).

[83] Shlegeris (2024).

# 4. Implications for Strategic Stability

Previous chapters have offered an overview of adversarial AI attacks and the potential attack surface, before exploring the possible impacts of prioritised threat vectors in more detail. Building on this foundation, this chapter identifies the overarching implications of adversarial AI for strategic stability, drawing on emergent themes from the illustrative scenarios and findings from the expert workshop and interviews.

### Box 8. Summary of findings: Chapter 4

**The key impact pathways through which adversarial AI attacks could affect strategic stability are:**

- The (perceived) vulnerability of military AI systems (i.e. AI-enabled C2, communications or rapid targeting) to adversarial attacks undermines the **credibility of defence to adversaries and allies.**
- Adversarial AI may tip the (perceived) **offence-defence balance** in favour of the offence. Asymmetrical investments in adversarial AI capabilities between states may further tip the balance towards offence.
- **Ambiguity** around attacks and their **attribution** creates opportunities for miscalculation and mistrust. Empowering non-state actors through democratisation of technology, specifically adversarial AI, further complicates attribution.
- **Entanglement of conventional and nuclear C3** increases the risk of an accidental attack on NC3, risking escalation.

Various factors **scale up or scale down the level of impact** of an adversarial AI attack, such as:

- Severity increases with the affected AI system's **degree of autonomy**.
- Severity increases for AI systems with **agency to plan, decide and act to achieve high-level objectives.**
- Severity decreases with the **ease and speed of detection** of the attack.
- Severity decreases with the **speed of resolution.**
- **Attribution** affects understanding of the attack in the broader operational context and informs possible reconciliation or retaliation options.
- Severity decreases with the **availability and quality of reversionary modes.**
- The **state of the world** (whether the attack occurs in peacetime or during an armed conflict) affects the criticality of the affected system, as well as the prioritisation of disruptions relative to other threats.

## 4.1. Impact pathways for strategic stability

Table 4.1 outlines the major implications of adversarial AI that the research team identified as particularly relevant and significant for strategic stability. The most acute instability risks and the deterrence aspects most stressed by adversarial AI are:

- **The (perceived) vulnerability of AI systems to adversarial attacks undermines the credibility of defence to adversaries and allies.** As AI becomes a larger part of defence capability, the perception that these systems or the underlying AI models could be compromised would weaken confidence in (conventional) deterrence among adversaries and allies. There is a risk that adversaries will perceive UK systems as vulnerable and authorise attacks, even when evidence suggests otherwise, because credible deterrence messaging is difficult to convey. Perceived vulnerability could also raise fears of corruption, penetration or extraction of critical datasets, algorithms and model weights, or doubts about security and reliability among allies, particularly when shared operations depend on robust, resilient interoperable systems.
- **Adversarial AI may tip the (perceived) offence-defence balance in favour of the offence.** Competing states may assess that adversarial AI capabilities tip the balance towards offence, as it appears easier to conduct attacks than defend against them – an imbalance that may disproportionately affect Europe and NATO given its defensive orientation, but which could embolden authoritarian or revisionist actors. Adversaries may feel they can gain strategic advantages without exposing personnel or equipment to physical harm. Asymmetrical investments in adversarial AI capabilities between states may further tip the balance towards offence. Crucially, however, the perception rather than the reality of a shifting balance is enough to be potentially escalatory.
- **Adversarial AI attacks may erode trust in AI systems and lead to hyper-caution.** Integrity-violation attacks that introduce unpredictability and uncertainty (whether in AWSs or in the behaviour of humans relying on decision-support tools[84]) increase the risk of miscalculation. More broadly, integrity-violation attacks on widely used LLMs risk destabilisation of society through the potential for information manipulation.[85] Especially when combined with kinetic attacks, large-scale misinformation campaigns can exacerbate the effects of the attack, e.g. by spreading doubt about the attribution of the attack and making it harder for heads of state to take appropriate action.
- **Ambiguity about the intent and the perpetrator of adversarial AI attacks increases the 'fog of war' and therefore poses escalation risks.** The difficulty of attributing these attacks can lower the threshold for actors to use them, increasing escalation risk. This is compounded by the fact that it is often hard to distinguish between 'normal' AI malfunction and deliberate adversarial attacks,[86] creating conditions for mistrust and strained relations. In addition, as adversarial AI technologies become more widely accessible, non-state actors gain greater capacity to conduct such attacks, making attribution even more difficult.[87] Equally, AI firms serving defence may be the victim, which adds complexity, as they may be commercially incentivised not to disclose breaches.
- **Entanglement of nuclear and conventional military systems risks inadvertent escalation**, either because an adversarial AI attack intended for AI-enabled conventional forces could unintentionally

[84] RAND Europe interview (January 2026).

[85] Molas (2024).

[86] RAND Europe interview (January 2026).

[87] Kaushik (2018).

affect NC3 or be interpreted as doing so. Adversarial attacks on AI systems used in NC3 threaten the secure second-strike capability. Additionally, while AI-enabled ISR is expected to reduce the effectiveness of hardening and concealment of nuclear weapons,[88] adversarial AI capabilities could negate these advantages or even undermine concealment, e.g. through privacy compromise attacks.

[88] van Hooft (2022).

Table 4.1. Examples of adversarial AI impacts on deterrence concepts; the symbols indicate whether this dynamic contributes to increased (▲) or decreased (▼) stability

| Strategic stability concept | Offensive incentive | | | | Mistrust | | |
|---|---|---|---|---|---|---|---|
| **Deterrence concept** | **Denying gains** | **Imposing costs** | **Offence/defence balance** | **Credibility** | **Threat to secure second strike** | **Attribution** | **Potential for miscalculation or mistrust** |
| **Implications of Adversarial AI** | ▼ Adversaries may perceive they can achieve strategic gains (e.g. damaging the opponent's military readiness or broader resilience) without risking their own forces. | ▲ The ability to target CNI may increase the ability to impose costs.<br><br>▼ Attacks on military systems (e.g. targeting) reduce the ability to impose costs. | ▼ Adversarial AI capability tips the (perceived) balance towards offence, as it is easier to conduct attacks than defend against them. This imbalance may disproportionately affect Europe and NATO given its defensive orientation.<br><br>▲ The ability to target AI-enabled AWSs, C2 and ISR systems could undermine an aggressor's ability to launch successful attacks, bolstering defence.<br><br>▼ Fears that an adversary may compromise the model architecture or weights for new military AI could create a 'use it or lose it' mentality. | ▼ The ability to undermine precision and other AI-enabled capabilities may decrease the credibility of conventional military forces, and credibility as an ally (e.g. in NATO) and beyond.<br><br>▼ Attacks on CNI or AI-enabled disinformation could undermine will-to-fight at the domestic level.<br><br>▼ Attacks on AI decision support tools, alongside AI-enabled disinformation, could undermine cohesion at the alliance level. | ▲ AI-enabled ISR is suggested to decrease hardening/concealment of nuclear weapons. Adversarial AI capabilities may negate this effect.<br><br>▼ Adversarial AI may undermine hardening or concealment of nuclear weapons by compromising privacy.<br><br>▼ Adversarial attacks on AI in NC3 threaten secure second-strike capability.<br><br>▼ Entanglement of conventional and nuclear C3 increases the risk of an accidental attack on NC3, risking escalation.<br><br>▼ States may perceive a strong first-strike incentive to stop an adversary from achieving AGI capability. | ▼ Difficulty of attribution may encourage adversarial AI attacks, which could lead to escalation.<br><br>▼ Empowering non-state actors through the democratisation of AI technology, specifically adversarial AI, further complicates attribution. | ▼ Difficulty distinguishing between random AI malfunctions and adversarial AI attacks may lead to mistrust or deteriorating relations.<br><br>▼ Adversarial AI attacks may erode trust in AI systems and lead to hyper-caution.<br><br>▼ Integrity violation attacks that increase AWS unpredictability or human DDS users increase the risk of miscalculation.<br><br>▼ Integrity violation attacks on broad-use LLMs risk destabilisation of society through the potential for information manipulation. |

Source: RAND Europe analysis.

## 4.2. Factors shaping impact of adversarial AI

The impact of adversarial attacks on AI systems is not a simple matrix-mapping exercise that depends solely on the attack type and the system affected. Many other contextual factors affect the severity of an adversarial AI attack's consequences, such as the affected system's autonomy, the ease and speed of detection, the speed of resolution and the availability of reversionary modes. The following paragraphs discuss each in more detail.

### 4.2.1. The severity of any destabilising effects at the strategic level increases with the degree of autonomy of the affected system

The degree of autonomy in a military AI system significantly shapes the severity of the implications arising from adversarial AI attacks. As human oversight decreases, both opportunities for adversarial manipulation and the potential scale of implications increase.[89] Table 4.2 summarises how different autonomy levels (human-in-the-loop, human-on-the-loop and human-out-of-the-loop) affect the system's exposure to attack and the severity of possible outcomes.

**Table 4.2. Implications of an AI system's level of autonomy for the attack surface and severity of impact**

| Autonomy level | Implication for the attack surface | Implication for the severity of impact |
|---|---|---|
| **Human-in-the-loop**<br>**A human is directly involved in the decision-making process and must approve or execute the final action. AI provides input or recommendations only.** | The communication channel between the human operator and AI adds an exposed attack surface; availability attacks can sever operator access or disable the system. | Lower autonomy limits immediate harm, but behaviour when human input is unavailable should be validated. |
| **Human-on-the-loop**<br>**AI operates autonomously within defined parameters, but a human monitors the system and can intervene or override if necessary.** | The communication channel between the human operator and AI adds an exposed attack surface; monitoring/override mechanisms can be targeted; delays or failures in human intervention may affect attack impact. | Medium severity: system may drift before detection; intervention may come too late in fast-moving contexts. |
| **Human-out-of-the-loop**<br>**AI functions fully autonomously, with no human input or oversight during attack execution.** | No human supervision to detect or correct adversarial manipulation. | Highest severity: AAI attacks may go undetected for longer, and consequences may escalate rapidly, especially in weapon systems. |

Source: RAND Europe analysis, building on GC REAIM (2025).

Another factor influencing this dynamic is the extent of corrigibility of the AI system when an adversarial attack is detected. Corrigibility is the extent to which an AI system can be corrected, modified or shut down by its operators without resisting, manipulating or deceiving them.[90] For highly corrigible systems, operators can interrupt and override the behaviour, while low-corrigibility systems may preserve the corrupted goals or resist shutdown.

[89] Global Commission on Responsible Artificial Intelligence in the Military Domain (2025).

[90] Soares (2015).

Additionally, the military user and/or commander's level of trust in the AI system affects how closely its recommendations are monitored and how much influence a model has over human decision making. Excessive trust may allow system drift to go unnoticed for longer, while insufficient trust can reduce effectiveness and introduce delays.[91] Adversarial attacks may also purposefully undermine trust: repeated false alarms (false positives) or missed threats (false negatives) can both erode confidence in the system over time.[92] This loss of trust negates the intended improvements in decision quality or speed that the AI system was designed to provide.

### 4.2.2. The severity of effects at the strategic level decreases with the ease and speed of detection of the attack

The speed with which an adversarial AI attack is detected and resolved strongly influences the nature and scale of its potential impact.[93] Early detection allows operators or contracted support to intervene, isolate affected components or shift to alternative decision-making processes (see the next subsection). Attacks that are harder to identify may influence outputs for longer before being noticed. Our expert workshop highlighted differences in how easily different attack types are detected. For example, an integrity violation attack may not become apparent until it is too late (i.e. systems begin malfunctioning). In contrast, limited- or denied-access attacks would, in most cases, be evident immediately and responded to rapidly. Response options to privacy compromise attacks, where (parts of) an AI system's algorithm, model weights or data are stolen, are limited, and countermeasures focus on prevention.

The expert workshop discussion also highlighted that almost any attack could have major impacts if left undetected long enough, including, for example, disruptions to procurement or hiring systems. Others might have immediate tactical or operational effects but could be quickly mitigated once identified, e.g. interference with logistics optimisation tools or disruption to shared civilian–military air-traffic control processes.

Building on lessons from the wider cybersecurity community, robust disclosure guidelines and reporting regimes would be required to ensure timely and prudent handling of any attacks on AI systems that occur across the wider defence ecosystem or AI lifecycle (e.g. when adversarial attacks target frontier AI companies or military AI firms, rather than the military end-user). Here, commercial incentives or other factors may discourage immediate disclosure of a detected adversarial AI attack, slowing any response.

### 4.2.3. The attribution of the attack affects its understanding in the broader operational context and informs possible reconciliation or retaliation options

Like any cyber-attack, attributing an adversarial AI attack to the ultimate responsible party may be extremely difficult because of anonymity tools, deliberate deception and proxies contracted directly or indirectly by state actors to conduct malicious activity on their behalf.[94] Politically motivated non-state hackers and private intermediaries acting on behalf of states blur the lines of responsibility, complicating efforts to

[91] Afroogh (2024).

[92] Apruzzese (2019); RAND Europe interview (January 2026).

[93] RAND Europe Expert Workshop (December 2026).

[94] Lin (2025); Mueller (2019); Nederveen (2024).

identify the attackers.[95] Sources for attribution include technical forensics, human intelligence, signals intelligence, history and geopolitics, among others.[96] While geopolitical circumstances can provide clues about who would want to launch a particular attack, this risks misattribution and, therefore, potentially misdirected mistrust, retaliation and inadvertent escalation.

Additionally, knowing that attribution is difficult may incentivise riskier operations, and increase the frequency of grey-zone operations that further erode mutual restraint. Furthermore, there is a risk of misunderstanding and misattribution of 'normal' system malfunctions, system drift or misclassifications.[97] Note that AI systems are, by definition, probabilistic, and therefore have a non-zero error rate. Mistaking these errors for adversarial attacks builds mutual mistrust and risks unintended escalation, while also making it difficult to achieve cohesion within a coalition or alliance on how to respond collectively.

### 4.2.4. The severity of strategic-level effect decreases with the availability and quality of reversionary modes

The workshop suggested that the availability and effectiveness of suitable alternatives for the compromised system's primary function affect the longer-term implications of adversarial AI attacks. The time required to either restore system functionality or revert to non-AI-enabled decision making and action is also critical. For example, a reversionary mode of an AI-enabled ISR system would be the (current) manual ISR process; traditional crewed platforms may be available to undertake tasks performed by a compromised AWS capability, if at greater risk to life. The expert workshop discussed that militaries already plan for scenarios in which technologies are compromised, where reversionary modes are a key element. These plans build resilience against adversarial AI attacks and other threats. Crucially, this means both technical and non-technical solutions, e.g. Tactics, Techniques and Procedures (TTPs), doctrine and training/exercising with reversionary modes as a core component to ensure forces are ready to employ them.

### 4.2.5. The state of the world (whether the attack occurs in peacetime or during an armed conflict) affects the criticality of the affected system, but also the prioritisation of disruptions relative to other threats

Finally, the state of the world in which an adversarial AI attack occurs affects the nature and scale of its implications. In a peacetime scenario, compromise of military AI systems would be considered to have strategic implications. In a wartime scenario, these implications would become operational by necessity.

Additionally, the duration of a conflict is a major factor in how an attack's implications and impacts materialise. For example, if adversaries steal a model for an autonomous weapon system, they may be unlikely to reproduce and field their own reverse-engineered system during a short war. However, they could do so over a longer conflict with more scope for force reconstitution and innovation. Similarly, the impact of attacks on recruitment would be less impactful in a short war but become critical in a long war. Workshop participants noted that it is uncertain whether adversarial AI threats would be a top priority in a wartime scenario, since adversarial AI attacks would only be one of a diverse range of threats.

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[95] Kaushik (2018).

[96] Lin (2016).

[97] UK Ministry of Defence (2022).

# 5. Governance Mechanisms and Responses

This chapter examines the policy and governance implications of adversarial AI. Building on the taxonomy of adversarial AI attacks, the assessment of potential targets and the mapped implications for strategic stability developed in earlier chapters, this chapter considers how existing national and international governance mechanisms do or do not address adversarial AI risks. It also outlines gaps that may require new or adapted governance responses and the challenges of establishing such robust mechanisms.

### Box 9. Summary of findings: Chapter 5

National and international policy documents acknowledge the vulnerabilities of military AI systems to adversarial attacks. For now, proposed solutions are primarily technical (e.g. Testing, Evaluation, Verification and Validation [TEVV], designing for robustness, and After-Action Reviews [AAVs]). Measures to increase resilience (i.e. ensuring rapid restoration of function) of affected systems, and explicit mechanisms to contain escalation risks, are underexposed. Political solutions to reduce escalation risk are limited to suggestions for information and incident sharing. While such soft governance approaches may suffice for confidence-building and signalling restraint, the absence of harder governance mechanisms poses acute strategic-stability risks amid possible adversarial AI threats to NC3 and dual-use C3 systems.

Although establishing more concrete, enforced mechanisms could enhance strategic stability, many challenges to tech governance exist in the current geopolitical environment, including the following:

- States have **varying appetites for automation and autonomy** in defence, and therefore binding rules risk being either overly restrictive for states that see autonomy as strategically necessary or too permissive for those seeking tighter boundaries.
- The **future trajectory of AI**, its integration into critical systems and the development of adversarial AI capabilities are all **uncertain**, which could make reactionary governance mechanisms obsolete.
- **No clear definition exists** for either the systems that should be under consideration or the extent to which automation should be acceptable, and thus where adversarial AI risks may accrue.
- While knowledge sharing and incident reporting are promising governance mechanisms to reduce the risk of misunderstandings, accidents or inadvertent escalation, states (or the commercial actors supporting them) may **hesitate to share vulnerabilities or disruptions** in their AI systems if doing so exposes them to further adversarial AI attacks.
- Treaty-based governance can be effective when states can **verify one another's behaviour,** but applying it to adversarial AI is far more difficult because AI systems are opaque, software-based and developed behind closed doors – and adversarial AI attack capabilities even more so.
- In strategic competition and a potential AI arms race, imposing comparatively strict regulation that **hinders innovation** may be perceived as contributing to falling behind in the AI arms race.

## 5.1. Existing governance mechanisms

Before examining governance mechanisms that apply to adversarial AI attacks, it is first useful to outline general approaches to governing (military) EDTs to reduce risk of escalation, maintain or improve alliance stability and avoid catastrophic outcomes. These governance mechanisms provide the framework and context for identifying opportunities and challenges in governing adversarial AI, as summarised in Table 5.1 below. Soft mechanisms are voluntary principles that guide behaviour but are not legally enforced. Hard mechanisms comprise binding rules that a court or other body can enforce. In this sense, the distinction concerns enforceability rather than effectiveness.

Table 5.1. Overview of possible governance mechanisms

| Type | Governance mechanism | Purpose |
|---|---|---|
| Soft norms and voluntary governance | Ethical norms, voluntary guidelines, codes of conduct | Establish expectations and approaches to the acceptable use of certain technologies. |
| | Bilateral and multilateral talks, summits and fora | Facilitate discussions and build consensus amongst states, organisations and experts. |
| | Information sharing arrangements | Agreements to corporation, increase transparency and build trust by sharing data. |
| | Independent partnerships (often between two states) | Collaborate on certain technology development, whether it be via sharing expertise or deploying new technologies together. |
| | Minilateral commitments | Establish shared values between a select number of states. Examples include the G7 and ASEAN. |
| | Resolutions and declarations by Intergovernmental Organisations (IGOs) | Formally outline and promote behaviours and standards. These are not legally binding unless ratified by domestic law. |
| | International standardisation bodies | Oversight and monitoring; responsible for setting standards to develop consensus. |
| | National or regional (e.g. EU) policies | Governments self-enforce policies until they enact them into domestic legislation, making them legally enforceable in court. |
| Binding instruments and rules | International treaty law (conventions) | Legally binding under international law, enforceable only against the signatories. |
| | Customary international law | General practice that has become accepted as international law. These obligations are binding across all countries and do not require ratification. |
| | Domestic law (e.g., export controls, investment screening) | Any legislation, regulation or by-law enforceable within the borders of a specific jurisdiction in which it was passed. |

Source: Bugos (2023); European Commission (2019); ICRC (2010); ICRC (2026); Maggetti (2015); Patrick (2015); Van den Rul (2016); The Policy Institute (2026); UK Government (2025); UNODA (2024); WEF (2019).

Ultimately, adversarial AI threats cut across ongoing governance discussions on AI, military AI, nuclear and cyber, and necessarily affect not only states but also a range of other stakeholders, including AI firms and civil society organisations. The sections below address each of these in turn.

This crosscutting of different thematic areas of governance poses both challenges and opportunities, implying that risk mitigations against the destabilising effects of adversarial AI attacks can build on and should become embedded in a variety of existing governance mechanisms and discussions, rather than necessarily requiring standalone initiatives or new stovepipes. At the same time, this risks reducing the prominence of the mounting risk from adversarial AI; as such, awareness-raising and capacity-building will be important to ensure adversarial AI issues are not overlooked in wider debates.

### 5.1.1. Military AI governance centres on normative commitments without binding oversight

This section outlines the key UK and international initiatives aimed at building norms and commitments around the responsible development, deployment and use of military AI. Because many of these efforts address military AI in broad terms, we consider AWSs, military decision-support tools and back-office applications together, highlighting system-specific issues where relevant.

While International Humanitarian Law (IHL) does not specifically regulate military AI, its principles of distinction (between civilians and combatants, and between civilian objects and military objectives), proportionality and precaution apply[98]:

- **Distinction:** AI systems must reliably separate civilians from military objects.
- **Proportionality:** AI outputs or recommendations must not cause commanders to attack disproportionately.
- **Precaution:** States must take 'constant care' to verify targets and minimise civilian harm, which extends to system design, testing and use of the AI system.

Due diligence on data integrity, model reliability, and vulnerability mitigation (e.g. poisoning, evasion, brittleness) follows from the precautionary obligation.[99] In other words, even without AI-specific rules, IHL requires states to take 'reasonable effort' (through testing, evaluation, verification and validation) to ensure military AI is reliable, robust and not prone to malfunction or adversarial attacks. Additionally, Article 18 of the Draft Articles on State Responsibility (DASR)[100] specifies that a state that coerces another state to commit an act is internationally responsible for that act 'if they understand the circumstances of the act.' For adversarial AI, this implies that an unintended targeting decision may be the legal responsibility of the state that conducted the adversarial AI attack.[101]

Various multilateral efforts are underway in an attempt to establish norms and common approaches to responsible military use of AI. Current governance of adversarial AI attacks in the military domain is

[98] Bruun (2025); Kwik (2024); RAND Europe interview (January 2026).

[99] RAND Europe interview (January 2026).

[100] International Law Commission (2001).

[101] Deeks (2026).

normative, non-binding and focused on risk management across the AI life cycle rather than enforcement or operational transparency. International and multilateral initiatives[102] (such as REAIM, the GC-REAIM Strategic Guidance,[103] principles suggested by the UN Group of Governmental Experts[104] and the Political Declaration on Responsible Military Use of AI[105]) converge on shared priorities: improving robustness against adversarial manipulation through rigorous testing and evaluation, cybersecurity safeguards against spoofing and hacking, continuous monitoring of system performance, and structured incident reporting and after-action reviews. These frameworks emphasise proactive design measures, data quality and standards, and ongoing verification to detect failures, loss of control or degradation over time, and encourage international knowledge sharing and confidence-building. However, governance remains fragmented and voluntary, with limited accountability or verification. Notably, neither the United States nor Russia participated in the recent third iteration of REAIM, held in Spain in early February 2026.

For its part, the UK lays out its risk management approach to adversarial AI in the Defence AI Strategy,[106] the policy paper 'Ambitious, safe, responsible: our approach to the delivery of AI-enabled capability in defence',[107] and the Joint Service Publication 'Dependable Artificial Intelligence (AI) in Defence' (JSP 936).[108] The UK focuses on managing adversarial risks through systematic testing, evaluation, and continuous review of AI-enabled systems, alongside strong cybersecurity and assurance practices. Key priorities include information sharing, developing joint and allied TEVV processes that explicitly consider adversarial attacks, providing comprehensive risk guidance for operators, and implementing confidence-building measures to mitigate escalation risks arising from system malfunction. Complementing this, the Government Communications Headquarters (GCHQ) and the National Cyber Security Centre (NCSC) contribute through standards development, ongoing threat analysis and lifecycle security planning, reflecting an approach that treats adversarial AI threats as an enduring operational and governance challenge requiring continuous oversight, adaptation and international collaboration. While the European Union addresses adversarial AI attacks in the EU AI Act, the Act does not apply to military AI systems.[109]

To summarise, multilateral and UK policy documents acknowledge the vulnerabilities of AI systems to adversarial attacks. For now, proposed solutions are primarily technical (e.g. TEVV, designing for robustness and After-Action Reviews). Political solutions to decrease escalation risk are limited to suggestions for information and incident sharing, and in any case primarily focus on military AI use or malfunction, not AI compromise or failure as a direct result of adversarial AI attacks.

[102] See Annex B for the full breakdown of these governance mechanisms.

[103] Global Commission on Responsible Artificial Intelligence in the Military Domain (2025).

[104] Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons System (2022).

[105] United States Bureau of Arms Control and Nonproliferation (2023).

[106] UK Ministry of Defence (2022b).

[107] UK Ministry of Defence (2022a).

[108] UK Ministry of Defence (2024a).

[109] European Union (2024).

### 5.1.2. Nuclear governance debates increasingly recognise risks from AI integration in NC3, but face broader political headwinds at a time of geopolitical tension

While no international rules currently impose explicit legal or regulatory restrictions on a state's NC3, the domain is not ungoverned. Existing legal obligations relating to nuclear weapons implicitly extend to emerging AI-related risks within NC3. For instance, under principles of state sovereignty, nuclear-armed states are responsible for the safe management of their arsenals and launch procedures.[110] These obligations shape how NC3 systems are organised and should inform any future integration of AI. Despite challenges surrounding secrecy and national security of nuclear-armed states, there is growing recognition of the need for more targeted international governance mechanisms to address the AI–nuclear nexus. Over the past four years, a range of diplomatic fora, international advisory bodies and national statements have begun calling for dedicated discussions on the risks posed by AI in the military domain, laying the foundation for similar discussions on integrating AI into the nuclear domain, with some explicitly highlighting the importance of international governance in this area.[111]

Beyond diplomatic recognition of these threats, the parties have also reached concrete agreements. Most notably, in November 2024 the United States and China agreed to a non-binding commitment to retain human decision making over nuclear launches. While Russia is not a party to this agreement, it is generally recognised that Russia holds the position that human oversight will remain in nuclear launch decisions.[112] The UK similarly states in its Defence AI Strategy that human control over the launch of nuclear weapons is a necessity, but does not go into much further detail.[113] These developments should, however, be understood within a broader assessment of the risks that AI technologies – and therefore also adversarial AI threats – pose in the nuclear domain. We separate the debate surrounding the governance of AI and the nuclear domain between a minimalist conception (focused narrowly on ensuring human control over launch decisions) and an extended conception (encompassing the wider set of risks introduced by AI-enabled systems across NC3 and related infrastructures):

- **Minimal conception:** Under the minimalist conception, the aim is to retain human agency over the last step in the chain, namely, the launch of nuclear weapons. In this area, there is clear agreement and, as evidenced by the US-China agreement, steps are being taken to ensure this remains the case. This commitment matters: by affirming that launch decisions will not be delegated to automated systems, states cannot use references to automated launch decisions to gain an escalatory advantage, such as threatening to grant dead-hand decision-making procedures to AI. Such signals would likely undermine other states' perception of their own security due to risks of mistakes or perceived increased chances of launch.
- **Extended conception:** The extended conception goes beyond safeguarding the final launch decision, seeking to limit and, more broadly, to de-risk AI integration across the broader nuclear infrastructure that informs, supports and enables nuclear launches, such as early-warning systems,

[110] Crawford (2024).

[111] Saltini (2025a).

[112] Saltini (2025a).

[113] UK Ministry of Defence (2022); Saltini (2023c).

> surveillance networks, secure communications and other components in the chain between data gathering and nuclear launch. This approach avoids a narrow focus on the last step of the process, and argues that ensuring leaders retain control over the launch decision is necessary but insufficient if AI errors, vulnerabilities or adversarial AI can distort the information shaping those decisions.[114]

Governance at the AI–nuclear interface remains cautious, incremental and largely non-binding, with a strong emphasis on preserving human control over nuclear decision-making while stopping short of comprehensive regulation. Since the 2020 NPT Review Conference,[115] key nuclear-armed states such as the UK, the United States and France have repeatedly affirmed commitments to human oversight for actions critical to nuclear weapons employment, establishing a baseline norm rather than enforceable constraints. Parallel efforts (including the UN Secretary-General's AI Advisory Board[116] and the REAIM Blueprint for Action[117]) reflect growing international recognition of AI-related risks, but engagement with NC3 has been indirect, politically sensitive and uneven, with language on human control contested and only selectively endorsed. Bilateral initiatives, notably the 2024 US–China agreement,[118] signal risk awareness but largely codify a minimal 'human-in-the-loop' standard. More recently, momentum has shifted modestly toward broader consideration of systemic risks, as reflected in the 2025 UN First Committee resolution calling for national policies that retain human control over NC3 and preclude autonomous nuclear launch decisions.[119]

These initiatives suggest that achieving agreement over the extended conception will be harder than the initial progress made in the minimal conception. The First Committee resolution is the first to acknowledge the extended conception as an important step in a governance system that adequately considers the full spectrum of threats that integrating AI into the nuclear domain brings to international security and escalatory dynamics. However, the resolution drew support from states that prioritise nuclear disarmament, as opposed to states that view disarmament as undesirable within the current international security environment, with the major nuclear weapons states of the United States, Russia, China, France and the UK, as well as their allies, ascribing to the latter view.[120] Despite the ideas found in the resolution, China, the United States and Russia may each be exploring the integration of AI into their NC3 systems.[121]

### 5.1.3. Governance of adversarial AI attacks on critical national infrastructure relies on existing international law and cyber norms, which are still maturing

IHL governs the targeting of critical national infrastructure during armed conflicts, since the infrastructure delivers essential services to the civilian population.[122] A piece of critical infrastructure becomes a military objective under IHL only if it both contributes directly to military action and if attacking it would give a

[114] Shanahan (2025).

[115] Saltini (2025a).

[116] Saltini (2023b).

[117] Rosen (2024).

[118] Mennesson (2025).

[119] Saltini (2025b).

[120] Saltini (2025b).

[121] Mennesson (2025); RAND Europe Interviews (January 2026); Saltini (2025b).

[122] Carlo (2024).

concrete, immediate military advantage. This requires a close, proximate link to military operations, such as supplying power to military facilities, command systems or war-production sites. IHL also bars attacks on energy infrastructure carried out mainly to hurt an enemy's economy, pressure negotiations, influence civilian morale or spread terror, and such illegitimate purposes may be inferred from patterns of indiscriminate or large-scale attacks on civilian infrastructure.[123]

Additionally, the Rome Statute of the International Criminal Court (ICC) states that attacks on infrastructure powering essential services (such as power, water or healthcare) come under Article 8 (war crimes) or Article 7 (crimes against humanity). The ICC also released a policy paper explaining how the Rome Statute can be interpreted in relation to non-human-controlled (e.g. cyber-enabled) actions.[124]

In peacetime, the UN's norms of responsible state behaviour in cyberspace apply – most notably norm (f) that states that a state should not conduct or knowingly support ICT activities that intentionally damage critical infrastructure or impair its operation.[125] However, the definition of critical infrastructure that falls under the norm is ambiguous. These frameworks are not consistently upheld, as illustrated by covert disruptions to subsea infrastructure. Cyber operations and adversarial AI attacks further complicate matters because attribution is often difficult, limiting accountability.

The UK's publication of the AI Security Code of Practice (which explicitly addresses adversarial AI attacks) led the European Telecommunications Standards Institute to develop a global standard of security requirements for AI systems.[126] As discussed in Chapter 2, standardisation bodies NIST and MITRE publish taxonomies and standards around adversarial AI.[127] Existing information-sharing partnerships likely include adversarial AI attacks, though they do not do so explicitly or separately.[128] While these codes of conduct and international standards help develop more robust AI systems, they do not reduce the escalation risk or build trust between non-allied states.

The EU addresses adversarial AI attacks in the EU AI Act, which applies to any non-military AI system deployed in or affecting persons located in the EU.[129] AI systems integrated in the management and operation of critical infrastructure are classified as high-risk, meaning they should adhere to more stringent diligence and oversight. The EU AI Act establishes mandatory risk management, robustness (including to adversarial AI attacks), cybersecurity and human oversight in pre-emption and response to AI system disruptions. The AI Act suggests developing technical solutions to address AI system vulnerability to adversarial AI, including 'measures to prevent, detect, respond to, resolve and control for attacks trying to manipulate the training data set (data poisoning), or pre-trained components used in training (model

[123] Giorgou (2023).

[124] International Criminal Court (2025).

[125] Haataja (2022).

[126] European Telecommunications Standards Institute (2025); Department for Science, Innovation and Technology (2025).

[127] MITRE Corporation (2026); Vassilev et al. (2025).

[128] Hitchens (2017).

[129] European Union (2024).

poisoning), inputs designed to cause the AI model to make a mistake (adversarial examples or model evasion), confidentiality attacks or model flaws.'[130]

## 5.2. Gaps

The review of existing governance of adversarial AI attacks in the military, nuclear and critical infrastructure domains is mapped to the framework of possible mechanisms. This results in the overview in Table 5.2, which summarises the extent to which existing governance does or does not address adversarial AI attacks.

**Table 5.2. Established governance mechanisms that address adversarial AI attacks in military AI, NC3 and CNI**

| Type | Mechanism | Military AI | NC3 | Cyber and CNI |
|---|---|---|---|---|
| Soft options & voluntary governance | Ethical norms, voluntary guidelines and codes of conduct | ✓ | ✓ | ✓ |
| | Bilateral and multilateral talks, summits and fora | ✓ | ✓ | ✓ |
| | Information-sharing arrangements | | | |
| | Independent partnerships | | | |
| | Minilateral commitments | | ✓ | |
| | Resolutions and declarations by IGOs | | ✓ | |
| | International standardisation bodies | | | ✓ |
| | UK national or regional policies | ✓ | | ✓ |
| Rules & incentives | International treaty law (conventions) | | | ✓ |
| | Customary international law | ✓ | | ✓ |
| | Domestic legislation (e.g. export controls) | ✓ | | ✓ |

Source: RAND Europe analysis.

Table 5.2 shows that recent discussions on military AI and related strategic systems show a strong emphasis on the 'softest' governance mechanisms. International fora currently prioritise non-binding norms, confidence-building measures and voluntary frameworks for military AI. Soft governance is likely adequate for confidence-building and signalling restraint, but the absence of harder governance mechanisms poses

[130] European Union (2024).

acute strategic-stability risks for NC3 and dual-use C3 systems. For critical national infrastructure, many of the commitments emerge from cybersecurity governance. Even when adversarial AI attacks are specifically mentioned, the main focus is on increasing robustness of the systems to prevent undesired outcomes. Measures to increase system resilience (i.e. ensuring rapid restoration of function) and mechanisms to contain escalation risks remain underexposed. Mechanisms to share information on incidents or 'hotlines' between governments are absent for military AI and NC3, despite the risks of unintended escalation through misinterpretation or misattribution of adversarial AI attacks. While more concrete, enforceable mechanisms could enhance strategic stability, many challenges remain.

## 5.3. Challenges for establishing governance around adversarial AI

### 5.3.1. The future trajectory of AI technology and pace of adoption is uncertain

Governments may hesitate to establish governance mechanisms for technologies whose future trajectory is uncertain. Reactionary treaties (as applied to, for example, chemical weapons) may be made obsolete by the speed at which technology changes.[131] At the other end of the spectrum, states may perceive technologically neutral governance as prohibitively over-inclusive,[132] preferring soft mechanisms that can co-evolve with the technology – as the UK, the United States and others are currently exploring and developing in frontier AI systems. Still, these lack the comprehensive stakeholder buy-in, scope or verification and enforcement mechanisms of hard governance mechanisms, creating trade-offs.

### 5.3.2. Different states have varying appetites for defence automation and autonomy

One reason hard governance for military AI is difficult to establish is that states have fundamentally different appetites for automation and autonomy in the use of force, shaped by their strategic cultures, threat perceptions and technological maturity. Some states view higher levels of autonomy as essential to maintaining military effectiveness, speed, and deterrence, while others emphasise human control because of ethical concerns or fear of escalation.[133] As a result, binding rules risk being either overly restrictive for states that see autonomy as strategically necessary or too permissive for those seeking tighter boundaries. States already have varying degrees of automation in their military operations, NC3 and critical national infrastructure, as well as different areas of automation. For NC3, there is an added challenge of heightened secrecy.

### 5.3.3. Governance of adversarial AI threats requires building agreed norms on what constitutes responsible (offensive) behaviour and responsible AI design

A challenge in discussing limiting the integration of AI into military systems, or regulating how states conduct or mitigate the risks from adversarial AI attacks, is the lack of a common, nuanced understanding or prioritisation of the issues of concern.[134] While there is broad agreement on cases such as the risks of AI

[131] Picker (2001).

[132] Maas (2019).

[133] Simonetti (2020).

[134] Saltini (2025b).

integration with NC3, many other use cases for AI in the military domain (and thus possible risks from adversarial AI attacks) remain more contested and ambiguous. No clear definition exists for which systems should be considered, the extent to which automation should be acceptable or what levels of assurance are needed to mitigate adversarial AI attacks. Governance debates would ideally draw a boundary to set clear limits on what can and cannot be attacked within a complex AI-enabled system of systems, but this poses a major challenge. Demarcating relevant areas of concern would be difficult when even unrelated applications, such as LLMs used for problem scoping and mission analysis, can affect the decision-making process (for example, on the situational awareness of those making nuclear decisions).[135] This holds for military AI, NC3 and CNI.

### 5.3.4. States and private firms may be hesitant to share vulnerabilities in AI systems

Knowledge sharing and incident reporting are often recommended as promising governance mechanisms to reduce the risk of misunderstandings, accidents, or inadvertent escalation.[136] The gold standard example often cited for these crisis communications mechanisms is the US-Russian Nuclear Hotline. However, such crisis communications mechanisms are more difficult to establish and sustain than assumed, as they depend on sustained political commitment, clear operating procedures and regular testing, which are weakened by mistrust and deteriorating relations. States may establish new, dedicated hotlines or incident-sharing mechanisms for, for example, autonomous weapons, and could help address incidents involving anomalous behaviour, malfunctions or adversarial exploits. However, states may hesitate to share vulnerabilities or disruptions in their AI systems if that exposes them to further adversarial AI attacks, and private firms developing AI technologies also have a clear commercial disincentive to share.

### 5.3.5. Verifying compliance with treaty-based governance is challenging due to the unobservability of AI capabilities

Treaty-based governing mechanisms can be highly effective when major actors agree to ratify and uphold them, creating clear obligations and shared expectations. However, applying this model to digital capabilities like adversarial AI attacks is challenging, as it is difficult to verify compliance or assess whether states are abiding by treaty commitments. Unlike, for example, nuclear arms control – where the physicality, complexity and visibility of nuclear weapons make them difficult to develop in secret and therefore amenable to monitoring and confidence-building measures – AI systems or the cyber tools for compromising them lack comparable observability. In contrast, AI is software-driven and typically developed behind closed doors, meaning traditional monitoring tools, including satellite surveillance, cannot reliably detect its development or intended use. Further, AI is integrated throughout society and is dual-use, meaning large-scale investments do not reveal its intended uses, further complicating monitoring.

[135] Shanahan (2025).

[136] Global Commission on Responsible Artificial Intelligence in the Military Domain (2025).

### 5.3.6. States may be hesitant to hinder innovation, given narratives of zero-sum competition and fears that an adversary may gain first-mover advantage in AI

Some state that regulation and governance stifle innovation. For example, some argue that current debates around AI integration into NC3 focus too much on threats and ignore the opportunities AI could contribute to international security.[137] For instance, NC3 infrastructure is already under strain due to ageing hardware, which increases the risk of mechanical or human error, while the growing technological sophistication of contemporary weapons systems threatens the resilience of these ageing systems.[138] Amid strategic competition over and through AI, hindering innovation by imposing comparatively strict regulation may be perceived to contribute to falling behind in a supposed AI arms race – even if this race exists more in the fears, distrust and misperceptions states have of each other than in reality.[139]

[137] Saltini (2025b).

[138] Crawford (2024).

[139] Khan (2025); Milmo (2025).

# 6. Conclusion

Based on the research and findings presented in this study, we recommend the following actions for the UK and like-minded nations, or wider private and civil society organisations, seeking to reduce the risks of strategic instability stemming from adversarial AI threats.

Collectively, these recommendations aim to:

- **Embed adversarial AI threats into relevant governance fora and mechanisms** for broader AI, nuclear and cyber, ensuring such threats do not fall through the cracks in international discussions.
- **Support awareness-raising and capacity-building around adversarial AI threats**, promoting responsible design principles and ensuring robust AI assurance frameworks that help militaries, governments and private firms to reduce the risk of models, data and outputs being compromised.
- **Decrease the risk of unintended escalation from adversarial AI** via multilateral norm-building.
- **Maintain alliance stability and deterrence** through information-sharing and the maintenance of credible AI-enabled defence capabilities.
- **Avoid catastrophic outcomes** by bolstering AI security and safety, sharing incident reports and encouraging restraint around targeting critical systems with adversarial AI attacks.

As such, the UK and like-minded nations have an opportunity to play a leading role in setting the agenda and shaping global practice on managing adversarial AI threats, as part of the broader debate around AI governance. The UK AI Security Institute has established itself as a leader in AI safety and security more generally, while the UK has governmental, private-sector and academic strengths in relevant fields such as AI assurance, red teaming, cybersecurity and nuclear. The FCDO has also led capacity-building on other technologies; for example, bolstering the cybersecurity of governments, armed forces and institutions in partner nations and emerging economies to help de-risk the potential for unintended escalation or conflict in the cyber domain.[140]

Recognising this as an exploratory study, and given the pace of change in both AI technology and the threat environment for adversarial AI, these recommendations are not intended to be comprehensive. Further research, analysis and dialogue will be required to iteratively refine the collective approach to managing the growing strategic stability risks associated with attacks on military and other AI systems.

[140] Black et al. (2024).

## 6.1. Recommendations

The following paragraphs set out our recommendations to the UK and like-minded governments, based on our analysis, interviews and expert workshop. Although AI is still being developed and integrated into military systems, meaning adversarial AI may threaten strategic stability only in the longer term, there is a real and current opportunity to proactively strengthen systems and steer (international) policy discussions.

### 6.1.1. Foreign ministries from concerned nations, such as the UK, should work to ensure fast-evolving adversarial AI threats do not slip through the seams between governance initiatives for wider AI, nuclear and cyber

Potential actions emerging from the analysis, interviews and expert workshop include the following:

- **Foreign ministries should ensure that governance debates and international fora tackling AI more broadly, as well as nuclear and cyber, embed adversarial AI in their discussions**, monitor the latest developments in adversarial AI tactics and technology and adjust accordingly. Selected examples include REAIM, work on LAWSs at the UN level, discussions around nuclear and the role of AI in NC3, and the ongoing development of a cyber–Tallinn Manual 3.0, all of which could benefit from the UK and like-minded nations foregrounding risks from adversarial AI.
- **Foreign ministries should promote after-action reviews** following significant adversarial AI incidents. These should assess the source, impact and systemic implications of attacks, and commit to taking steps to prevent recurrence. Adversaries as well as allies could be interested in establishing a catalogue or forum for sharing insights.
- **Foreign ministries should promote the inclusion of system resilience considerations in multilateral governance mechanisms.** While both multilateral and UK policy documents acknowledge the vulnerabilities of military AI systems to adversarial attacks, proposed solutions remain largely technical for now (e.g. TEVV, designing for robustness and After-Action Reviews). These technical solutions primarily aim to reduce the likelihood of adversarial AI attacks. Measures to increase system resilience (i.e. ensuring rapid restoration of function) and mechanisms to contain escalation risks remain underutilised. Political solutions to reduce escalation risk are limited to suggestions for information sharing and incident sharing.
- **Allies and partners should ensure regular information sharing** on adversarial AI techniques and mechanisms to build confidence in incident detection, analysis and attribution. This could include horizon scanning and foresight activities to understand possible new threat vectors or the implications of changing AI and adversarial AI technologies.
- **Nations with more mature AI ecosystems should work with emerging economies to raise awareness of adversarial AI threats and build institutional capacity to mitigate risks domestically,** e.g. by sharing best practice on responsible AI design, incident management and reporting, or how to implement robust AI assurance mechanisms in defence and security organisations.
- **Foreign ministries should continue to work to shape emerging international norms around adversarial AI and what constitutes responsible behaviour in the deployment, compromise or targeting of different AI systems in a defence context,** building, for example, on the recent US-

China bilateral agreement on AI and nuclear. While agreement on comprehensive norms (let alone treaty-based mechanisms) is unlikely, progress is more feasible in areas of clear common interest, e.g. avoiding escalation risks around AI in NC3 or misuse of AI to proliferate bioweapons. Even though the UK has no intention of exploiting AI in any part of its NC3 system, it can put these topics on the agenda and work with other nuclear-armed states to minimise risk. Wider transparency and confidence-building measures (e.g. publishing relevant policy, doctrine and findings from adversarial AI incidents, etc.), as well as Track 1.5 and 2 dialogues or activities such as table-top exercises and simulations, could build common understanding of priorities and areas of mutual concern when it comes to the risks from adversarial AI attacks to strategic stability.

- **Foreign ministries should engage with competitors and adversaries to establish mechanisms to support timely communication on adversarial AI incidents**, for example through a Direct Communication Line, to de-escalate (suspected) adversarial AI attacks. While challenging to establish, this could contribute to reducing the risk that such attacks are misunderstood or misattributed, build longer-term confidence and support the gradual development of shared expectations around responsible behaviour in adversarial AI. Nations such as the UK could lead in this effort.

### 6.1.2. Wider government should continue to improve AI system security, robustness and resilience, reducing the risk of catastrophic outcomes and maintaining credible AI-enabled defence and deterrence to manage escalation risk

Potential actions emerging from the analysis, interviews and expert workshop include the following:

- **Defence ministries, security services and AI Security Institutes (or national equivalents) should collaboratively invest in building national capability and expertise on adversarial AI and the threats in a defence and security context.** Existing expertise is currently dispersed across government organisations and agencies, industry and academia. Insights should feed across wider government (e.g. MOD, FCDO, DSIT, Cabinet Office, AISI and others in the UK context). Working with industry and academia, dedicated teams could explore adversarial attack techniques and test military AI models at the system level, improving understanding of real-world vulnerabilities and failure modes. This capability is essential to inform proportionate governance, improve operational resilience and support credible international engagement.
- **Defence ministries should continue investing in robust military AI systems and rigorous assurance frameworks.** Military AI systems should be designed to resist adversarial manipulation across their entire lifecycle, including input data sanitisation, rigorous model validation, adversarial training, ongoing vulnerability analysis and demonstration of corrigibility. Technical mitigation frameworks, such as those catalogued by MITRE ATLAS, provide a valuable reference and should be incorporated into development and assurance processes where appropriate. In line with the distinction made in the EU AI Act, governments and defence ministries should hold military (and NC3 in particular) and CNI AI systems to higher standards around robustness and resilience than commercial or regular business applications.

- **Nations should develop, first at a cross-government level and then internationally with allies and partners, a clear position on acceptable levels of adversarial AI attacks, defined red lines, and corresponding courses of action.** This position must then be clearly and effectively communicated to adversaries, as would be done for physical or kinetic activity a nation seeks to deter. Until the fundamental uncertainty around the response to adversarial AI attacks is resolved, nations may struggle to respond effectively to sub-threshold adversary activity, or develop and articulate shared norms of responsible behaviour in this regard.
- **Governments should invest in building resilience into the broader systems supported by AI.** As the cyber landscape remains offence-favoured (it is often easier to mount attacks than to defend against them), the wider system needs to be conscious of the potential vulnerabilities and risks. Robustness at the AI model level is insufficient on its own. Resilience must be built into the wider system within which AI operates. Building resilience includes considering how failures propagate, ensuring the availability and quality of reversionary modes (e.g. alternative technologies, manual processes, and appropriate doctrine, TTPs and training), and considering how quickly systems can recover after an attack. In defence, parallel systems should be developed where feasible to avoid single-point failures, and mitigation and recovery plans should be established, tested and regularly exercised. At the national level, governments should include these considerations in their requirements when developing or procuring AI systems for critical functions or CNI.

## 6.2. Areas for further research

Given the limited scope and duration of this exploratory study, the findings and recommendations in this report are intended as a basis for further scrutiny, debate and analysis. The RAND team identified potential opportunities for additional research to inform collective understanding and action on de-risking the growing threats from adversarial AI to strategic stability, including:

- **Horizon scanning** and foresight into possible future developments in adversarial AI threat vectors.
- **Deep dives or comparative analysis** of the evolving perspective of different international actors (e.g. China and Russia) on adversarial AI and its governance.
- **Analysis of the influence of adversary strategic culture** on evolving doctrine and approaches to the preparation and conduct of adversarial AI attacks against military or non-military AI systems.
- **Serious games or table-top exercises** to explore specific scenarios and use cases in detail, possibly as part of a wider Track 1.5 or Track 2 dialogue to build common understanding internationally.
- **Further research into methods for improving the detection of both cyber and physical unfamiliar adversarial AI attacks**, and accounting for how multiple AIs interact in a system of systems.
- **Further research into methods for bolstering the contribution of humans to early detection and effective management of adversarial AI attacks** on systems they interact with or oversee, to inform development and rollout of relevant training, education and exercising.

- **Further research into operationalising avoiding catastrophic outcomes**, both when making one's own systems more robust and when thinking about offensive adversarial AI capabilities.
- **Further research into lessons from broader deterrence theory for adversarial AI,** including lessons on nuclear, cyber and cross-domain dynamics, and the potential to avoid destabilising first-strike incentives to launch adversarial AI attacks.
- **Development of a capacity or maturity model** for assessing and benchmarking the rigour of different organisations' risk management frameworks for dealing with adversarial AI threats to their AI systems and the potential impact on their broader operations (e.g. to support capacity building with defence ministries, governments or private-sector actors).
- **Continuing research into preparedness and resilience measures** to avoid adversarial extraction of advanced AI models and AGI to prevent loss of control.

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# Annex A. Methodology

This annex provides an overview of the mixed-methods research approach employed in this study.

## A.1. Literature review

The research team reviewed academic and grey literature to develop a taxonomy of adversarial AI threats and identify potential targets whose disruption could plausibly affect strategic stability, such as critical financial, government, energy, and military infrastructure. For this, the research team designed the following Boolean search string:

> ("adversarial machine learning" OR "adversarial ML" OR "adversarial AI" OR "adversarial artificial intelligence") AND (“taxonomy” OR “typology” OR “types”)

Applying the search string to PrimoSearch[141] resulted in 508 unique sources published from 2019 onwards, of which 11 were relevant to developing the taxonomy of adversarial AI threats, and 50 to identifying potential targets. We used a structured data extraction approach to extract relevant information from these sources, including the used or proposed taxonomy of adversarial AI attacks, the mentioned attack surface, and the implications for strategic stability. We incorporated the extracted information into a narrative that informed the taxonomy of adversarial AI used throughout this report and a longlist of AI systems that may be vulnerable to adversarial AI attacks.

## A.2. Expert workshop

In addition to internal analysis and desk-based research, we organised an external half-day workshop with a mix of experts and stakeholders to identify the implications of adversarial AI threats on strategic stability and how these vary by target(s) or context. For each potential target, the implications of adversarial AI threats were identified based on the taxonomy. Table A.3 contains the long list of targets discussed in the workshop, before summarising the findings into the five broader categories discussed in this report (see Chapter 3).

[141] PrimoSearch is a library collection search tool that allows for searching through articles, eBooks, print books and research reports. It includes academic databases like Scopus, Web of Science and arXiv, as well as grey literature sources like the Homeland Security Digital Library and US Declassified Documents Online.

Table A.3. Functions where AI could be integrated, which could subsequently become targets of adversarial AI attacks, as discussed during our expert workshop

| Category | Function where AI could be integrated |
|---|---|
| Autonomous systems: firepower | Object detection |
| Autonomous systems: firepower | Targeting |
| Autonomous systems: firepower | Navigation |
| ISTAR | Live data monitoring |
| Sustainment | Logistics optimisation |
| Training | Synthetic training environments |
| Back-office | HR systems |
| Back-office | Procurement |
| Back-office | Back-office process optimisation (incl. software development and use of CoPilot-like tools) |
| Back-office | Anomaly/fraud/malware/cyberattack detection |
| Critical national infrastructure | Financial systems |
| Critical national infrastructure | Energy systems |
| Critical national infrastructure | Drinking water supply |

Participants were guided by steering questions (Box 11) which considered the potential strategic, operational and tactical implications of adversarial AI attacks for strategic stability. The workshop was structured in two phases: Phase 1 considered implications in the contemporary world, while Phase 2 considered implications in a future conflict.

The workshop took place virtually on MS Teams and utilised Mural as a data-capture tool. The workshop brought together relevant national and international AI security experts, strategic stability experts and policymakers, along with the RAND team. The selection of interview candidates followed FCDO guidelines on gender equality and social inclusion (GESI).

Based on the workshop output, the research team created a matrix that distinguished the strategic impacts of different adversarial AI threats on targets identified during the literature review. The team then used this matrix to evaluate governance of adversarial AI threats.

### Box 11. Guiding questions used for the expert workshop

1. Strategic level:
    a. Would an attack compromise the credibility of deterrence?
    b. Could an attack lead to escalation (either deliberate or inadvertent)?
    c. To what extent might an adversarial AI capability shift the offence/defence balance?
    d. How would an attack impact resilience to shocks/surprise?
    e. How likely is an attack to challenge legal and ethical norms?
2. Operational level:
    a. Do attacks affect the operational level (or more the tactical)?
    b. Would a compromised AI system fundamentally change how operations are planned and conducted?
3. Tactical level:
    a. Would the AI system be controlled by tactical commands?
    b. How might an attack impact the six tactical functions (firepower, mobility, protection, sustainment, ISTAR and C2)?

### Table A.2. List of participants at the expert workshop

| Name | Position and organisation |
|---|---|
| Aishwaryaprajna | Lecturer in Computer Science at University of Exeter |
| Alice Saltini | Researcher at James Martin Center for Nonproliferation Studies; Arms Control Negotiation Academy; Centro Interdisciplinare Scienze per la Pace; Rhode Island School of Design |
| Anna Nadibaidze | Researcher at the Center for War Studies, University of Southern Denmark |
| SME on AI governance * | Strategy at the Ministry of Defence: Defence AI and Autonomy Unit |
| SME on AI governance * | Emerging and Disruptive Technology at FCDO |
| SME on military AI * | Ministry of Defence: Defence AI and Autonomy Unit |
| SME on military AI * | - |
| SME on military AI * | - |
| SME on AI governance * | - |

*These participants expressed a wish not to be (fully) identified in the report.

## A.3. Key informant interviews

The research team conducted a series of semi-structured interviews with subject matter experts to build upon the comparative literature review. Interviews lasted between 45 minutes and an hour, and combined a set protocol (Box 10) with scope to ask follow-up questions based on given responses.

Interviewees were AI governance experts drawn from governmental departments, as well as (academic) researchers from relevant organisations. The selection of interview candidates was informed by gender equality and social inclusion (GESI) concepts following FCDO guidelines. All contributors were asked to fill out a consent form and given options for different levels of attribution or anonymisation.

**Box 10. Topic guide used for the interviews**

1. What do you think are the main types of targets of adversarial AI attacks, and which of these could have the largest impact on the security of the UK or on the geopolitical context?
2. Aside from those already discussed, are there any other key potential targets of adversarial AI that we have not discussed?
3. What are the factors that influence the impact of an adversarial AI attack?
4. How do you think adversarial AI affects strategic stability?
5. The literature review has identified a taxonomy of adversarial AI attacks and, in discussion with the FCDO, a list of potential civil and military targets of adversarial AI attacks. A workshop with subject matter experts, consisting of relevant (external) national and international AI security experts, strategic stability experts and policymakers, has mapped the implications of adversarial AI threats on potential targets on strategic stability. For each of these scenarios:
    a. What do you think is the appropriate governance response, and the UK's role therein?
    b. Do appropriate fora/platforms (including for Track 1.5 or 2 diplomacy) exist to address these threats?
6. Can you highlight or identify any gaps in the governance of adversarial AI attacks?
7. What lessons can we learn from regulatory/governance approaches in other sectors, such as emerging cyber norms, in the development or establishment of new norms?

Table A.2. List of interviewed experts

| Name | Position and organisation |
|---|---|
| David Sully | CEO and co-founder at ADVAI |
| Nehal Bhuta | Professor of Public International Law at the University of Edinburgh |
| Alice Saltini | Researcher at James Martin Center for Nonproliferation Studies; Arms Control Negotiation Academy; Centro Interdisciplinare Scienze per la Pace; Rhode Island School of Design |
| Lt. Gen. John N.T. Shanahan | Former director of the Chief Digital and Artificial Intelligence Office |

## A.4. Governance mechanisms and gap analysis

The research team conducted the following research steps to map existing governance mechanisms, identify gaps and challenges to establishing governance, and formulate recommendations.

- **Initial mapping of governance mechanisms.** The research team conducted a structured mapping exercise using targeted searches of relevant policy papers, ensuring consistency across the programme of work (including the research on novel effectors and C2). The research team undertook additional desk research to identify existing governance approaches and challenges to establishing governance – particularly for AI and adversarial AI.
- **Targeted interviews with subject matter experts.** The preliminary findings were tested with key experts on adversarial AI and AI governance.
- **Matrix exercise to identify governance gaps.** The identified established governance mechanisms were structured on a scale from soft to hard governance approaches. This exercise revealed gaps in the governance landscape.

# Annex B. Analysis of Governance Mechanisms

Various multilateral efforts attempt to establish norms and common approaches to the responsible use of AI by the military. Relevant examples include:

- **The Global Commission on Responsible Artificial Intelligence in the Military Domain (GC REAIM) Strategic Guidance Report[142]:** The GC REAIM Strategic Guidance Report recommends several practices to address disruptions to AI systems, including:
    1. Knowledge Sharing: The report recommends promoting international-level knowledge sharing between civilian AI governance bodies and industry. Structured mechanisms could support this, for example via the UN Independent International Scientific Panel on AI and the Global Dialogue on AI Governance.
    2. Incident Reporting: The report recommends facilitating internal reporting of incidents, malfunctions or adversarial events and enabling analysis across operational teams.
    3. Developing standards: The report recommends developing data quality requirements, specifications, and standards for training, post-deployment learning, protection, and decommissioning. This includes reviewing datasets to identify unwanted biases and ensuring robustness against adversarial manipulation and data spoofing.
    4. Continuous Testing, Evaluation, Verification and Validation (TEVV): The report recommends ensuring continuous understanding of system performance, including data drift, limitations, failure modes and associated risks, to preserve authority and control of the system.[143] The report further advises conducting TEVV more frequently for systems that learn or update during real-world use.
    5. Developing requirements for robust and proactive After-Action Reviews (AARs): The report recommends conducting regular AARs to assess system performance and identify lessons learnt and areas for improvement.
- **Joint submission to the United Nations Group of Governmental Experts on Lethal Autonomous Weapon Systems on 'Principles and good practices on emerging technologies in the area of lethal**

[142] Global Commission on Responsible Artificial Intelligence in the Military Domain (2025).

[143] Allik (2024).

**autonomous weapons systems'[144]:** This proposal, submitted by Australia, Canada, Japan, the Republic of Korea, the United Kingdom and the United States, recommends rigorous testing and evaluation of AI systems for operational environments, cyber-security safeguards against spoofing or hacking, and consideration of system control loss risks during the entire life cycle.

- **Guiding Principles affirmed by the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems (GGE on LAWS):** These 11 Guiding Principles recommend developing appropriate (non)physical safeguards against hacking and spoofing, and pre-emptively integrating risk assessment and mitigation measures in system design, development, testing and deployment.
- **Political Declaration on Responsible Military Use of Artificial Intelligence and Autonomy[145]:** Initiated by the United States, the political declaration states that the signatory states will ensure rigorous and appropriate testing to ensure the safety, security and effectiveness of military AI; monitor systems to ensure safety features are working as intended; and implement safeguards, including the detection of disruptions and the deactivation or disengagement of deployed AI systems.

The UK lays out its risk management approach to adversarial AI in the Defence AI Strategy, the policy paper 'Ambitious, safe, responsible: our approach to the delivery of AI-enabled capability in defence', and the Joint Service Publication 'Dependable Artificial Intelligence (AI) in Defence' (JSP 936).[146] The UK focuses on managing adversarial risks through systematic testing, evaluation, and continuous review of AI-enabled systems, alongside strong cybersecurity and assurance practices. Key priorities include information sharing, developing joint and allied TEVV processes that explicitly consider adversarial attacks, providing comprehensive risk guidance for operators, and implementing confidence-building measures to mitigate escalation risks arising from system malfunction. Complementing this, the Government Communications Headquarters (GCHQ) and the National Cyber Security Centre (NCSC) contribute through standards development, ongoing threat analysis, and lifecycle security planning, reflecting an approach that treats adversarial AI threats as an enduring operational and governance challenge requiring continuous oversight, adaptation, and international collaboration.

- **UK Defence AI Strategy[147]:** The UK Defence AI Strategy considers both the risks and opportunities associated with autonomous weapon systems and military decision-support systems. It emphasises the importance of information sharing with AI developers via permission and declassification of data mechanisms; highlights aims to establish a cross-domain joint warfare centre to tackle AI-related operational challenges; recommends developing joint TEVV processes that consider adversarial AI attacks with international allies to facilitate the sharing of best practices and lessons

[144] Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons System (2022).

[145] United States Bureau of Arms Control and Nonproliferation (2023).

[146] UK Ministry of Defence (2024b).

[147] UK Ministry of Defence (2022b).

learned; the development of comprehensive risk guidance for operators; and establishing confidence-building measures to mitigate the risk of escalation in instances of system malfunction.

- **Policy paper 'Ambitious, safe, responsible: our approach to the delivery of AI-enabled capability in defence'[148]:** This policy paper establishes the mission to establish suitably reliable military AI systems, including regular review and testing of the reliability, robustness and security.
- **GCHQ's report 'Pioneering a New National Security: The Ethics of Artificial Intelligence'[149]:** This report outlines the UK's collaboration with the European Telecommunications Standards Institute in developing AI assurance and security standards and the efforts of the NCSC in analysing ongoing cyber threats to develop novel methods for countering AI system vulnerabilities. Additionally, GCHQ highlights maintaining lifecycle plans for each adopted AI system to ensure long-term security.

The following notable initiatives specifically govern the integration of AI into NC3:

- **2020 Review Conference of the Parties to the Treaty on the Non-Proliferation of Nuclear Weapons:** A working paper submitted by France, the UK and the United States outlined a commitment to retaining human oversight and involvement 'for all actions critical to informing and executing sovereign decisions concerning nuclear weapons employment.'[150]
- **UN AI Advisory Board:** Established by the UN Secretary-General in October 2023, the Advisory Board aims to build a shared understanding of AI-related risks and develop initial recommendations for international AI governance. While this does not include NC3, its outputs could act as a reference point for discussions at more nuclear-specific fora, such as the NPT.[151]
- **The REAIM Blueprint for Action:** This non-binding document outlines a series of principles on AI in the military domain, with a focus on specified areas including the nuclear domain.[152] Initial references to human oversight over the nuclear domain were removed from the document, reportedly because of some countries' concerns that this legitimised nuclear weapons.[153] However, in September 2024, a further refined version included the sentence 'it is especially crucial to maintain human control and involvement for all actions critical to informing and executing sovereign decisions concerning nuclear weapons employment.'[154] The document was led by the United States and supported by nuclear powers, such as the UK, France and Pakistan, but not China – though China was present at the Summit.[155]

[148] UK Ministry of Defence (2022a).

[149] Government Communications Headquarters (2021).

[150] Saltini (2025a).

[151] Saltini (2023b).

[152] Rosen (2024).

[153] Saltini (2025a).

[154] Saltini (2025b).

[155] Saltini (2025b).

- **US-China Bilateral Agreement:** Signed in November 2024, this non-binding agreement between the two major powers recognises the risks at the AI-nuclear interface. Despite being an important step, the agreement does not expand beyond the minimal 'human-in-the-loop' baseline.[156]
- **First Committee Resolution:** In November 2025, Austria, El Salvador, Kazakhstan, Kiribati, Malta and Mexico pushed through a text that led to a resolution within the UN General Assembly's First Committee, which is dedicated to discussions of disarmament and security.[157] Significantly, the text represents an initial step away from a minimal conception toward a broader nuclear architecture, explicitly considering the broader risks surrounding AI integration into NC3.[158] The resolution calls on states to adopt and publish national policies and doctrines that retain human control over NC3 and that autonomous systems will not initiate decisions on the use of nuclear weapons. The resolution received broad support, with 115 states voting in favour, 8 voting against and 44 abstaining.[159]

[156] Mennesson (2025).

[157] Saltini (2025b).

[158] Saltini (2025b).

[159] Saltini (2025b).